

SMRs without water

the sodium-cooled fast reactor

Sodium properties · metal fuel · Toshiba 4S · the 1986 EBR-II tests · a sodium system code

Weeks 5-6 · Instructor **Minseop Song** · Rev1.0 · September 2026

How Part 2 is built - principle, fuel, systems, safety, system code

- Part 1's HTGR bought passive safety by **lowering** power density. The SFR aims for it while **raising** power density to hundreds of MW/m³
- The evidence is two tests at EBR-II on 3 April 1986 - chapter 10 is the centre of this deck

07 Sodium

the only practical coolant that carries heat without moderating

08 Metal fuel

lower melting point - so why safer?

09 4S and the pool layout

what a 30-year no-refuelling reactor looks like

10 SFR safety

the reactor that shut itself down in 1986

11 Sodium system code

easier than helium, and harder in places

07

Why use a metal that explodes with water

The SFR starts from the neutron, not the coolant. Keeping neutrons fast means the coolant must not moderate. That one condition narrows the field to sodium, and each property of sodium then fixes one part of the design.

Why a fast spectrum, and the coolant candidates

- Fast neutrons can **breed and burn U-238 as plutonium** and **fission the transuranics away** - but only if the coolant does not moderate
- Exploding with water was not reason enough to give that up - it was **designed around**, and the detour is **the intermediate loop** (ch. 9)

Water

Out

moderating is its job

Helium

Out

no moderation, but cannot carry hundreds of MW/m³

Lead · Pb-Bi

Possible

heavy, barely moderates;
melts at 327 °C, corrodes steel

Sodium

Chosen

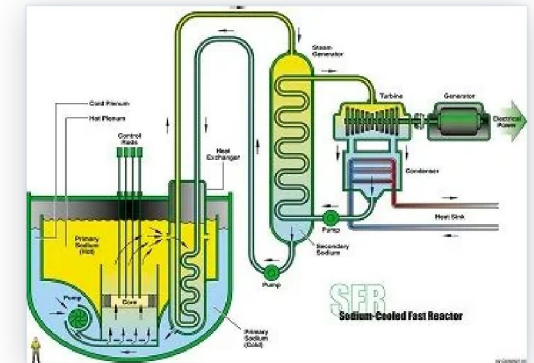
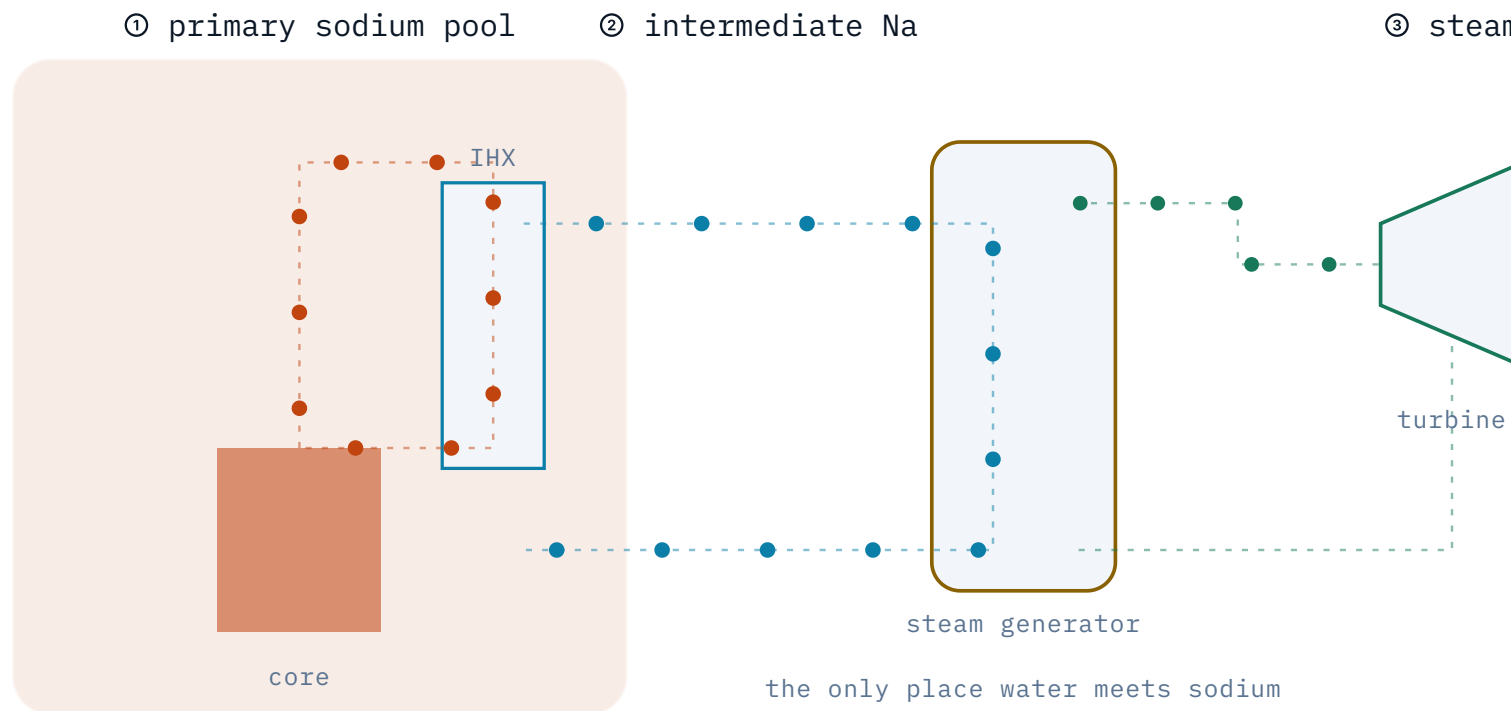
weak moderation;
conductivity 70 W/(m·K)

fast neutron an unmoderated, high-energy neutron · **transuranics** elements heavier than uranium · **scattering cross-section** the likelihood of a neutron bouncing off · **lead, lead-bismuth** other liquid-metal coolants

Source This textbook §71 · U-238 is 99.3 % of natural uranium · the transuranics (Np, Am, Cm) barely fission with slow neutrons and last 100,000 years

The pool-type SFR - primary, intermediate and steam circuits

- Primary sodium never leaves the pool. **Intermediate sodium** takes the heat to the steam generator - only it ever meets water
- Only the SFR has three circuits. Every practical SFR accepts an extra exchanger, pump, piping and tens of kelvin lost (end of ch. 9)



Pool-type SFR concept · textbook Fig. 7.1

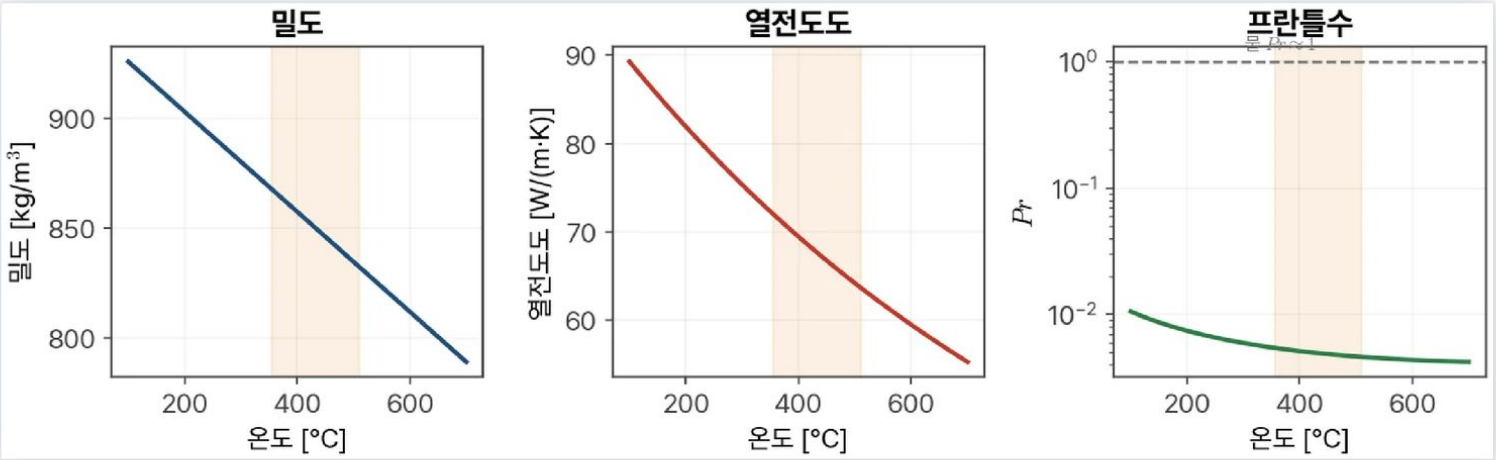
PRIMARY · ACTIVATED NA
Na-24 · 15 h half-life

pool type core, pumps and IHX all in one sodium tank · **IHX** intermediate heat exchanger · **intermediate loop** a non-radioactive sodium circuit between primary sodium and water

Source This textbook §7.1, Fig. 7.1; §9.6 · primary sodium (pool), intermediate sodium, steam-water

Sodium properties - the standard fits and the unit trap

- Linear fit for 100-600 °C: $\rho = 948.6 - 0.228 T(^{\circ}\text{C})$, within 0.1 %
- The common $\rho = 1014 - 0.235 T$ holds **with T in kelvin** (400 °C → 856 kg/m³). Put in Celsius and it is 7 % high - Volume III's Sodium() is in kelvin too



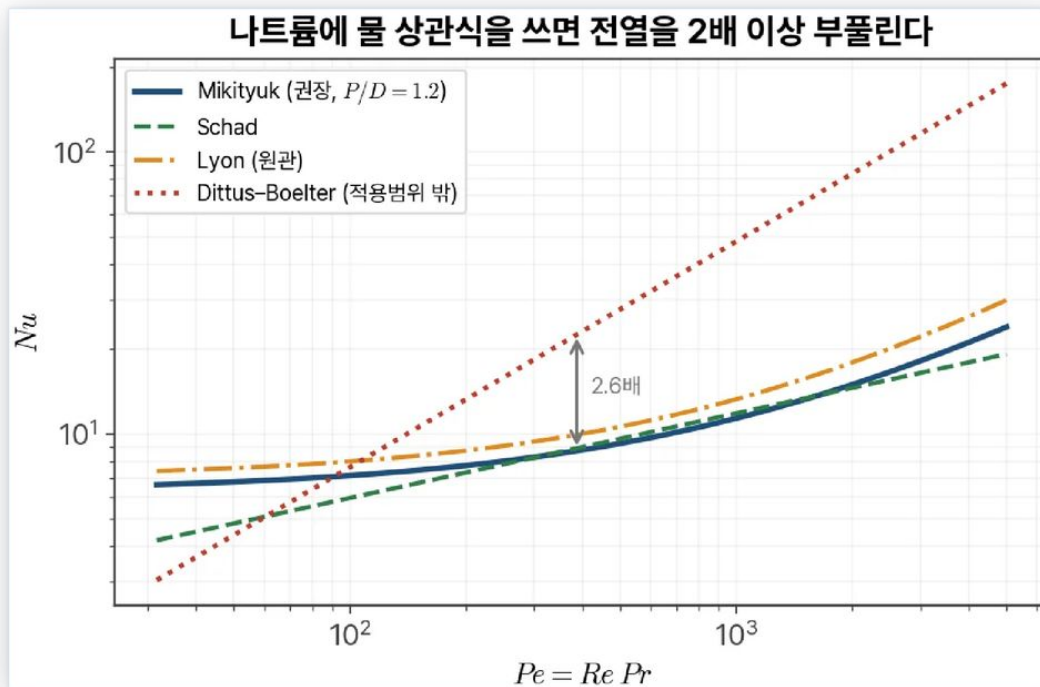
T °C	P	C _p	K	PR
100	925.2	1382	89.3	0.0105
355 (4S in)	868.0	1294	72.0	0.0054
400	857.7	1283	69.5	0.0051
510 (4S out)	832.3	1263	63.7	0.0046
883 (boils)	742.5	1271	48.6	0.0041

ρ kg/m³ · c_p J/(kg·K) · k W/(m·K)

Density, conductivity and Prandtl number of sodium - shading is the 4S range · textbook Fig. 7.2

$Pr \approx 0.005$ - liquid-metal heat transfer and the Mikityuk correlation

- Heat spreads **200 times faster** than momentum - molecular conduction carries much of it even in the turbulent core, so **the basis of Dittus-Boelter collapses**
- Everything collapses onto **$Pe = Re \cdot Pr$** . As $Pe \rightarrow 0$, Nu **stays finite** - the physics of conduction without flow is built in



Liquid-metal correlations against Dittus-Boelter - a water correlation inflates sodium heat transfer more than twofold · textbook Fig. 7.3

$$Nu = 0.047 [1 - e^{-3.8(P/D-1)}] \times (Pe^{0.77} + 250)$$

SODIUM Pr (400 °C)

0.0051

BORISHANSKII'S WARNING

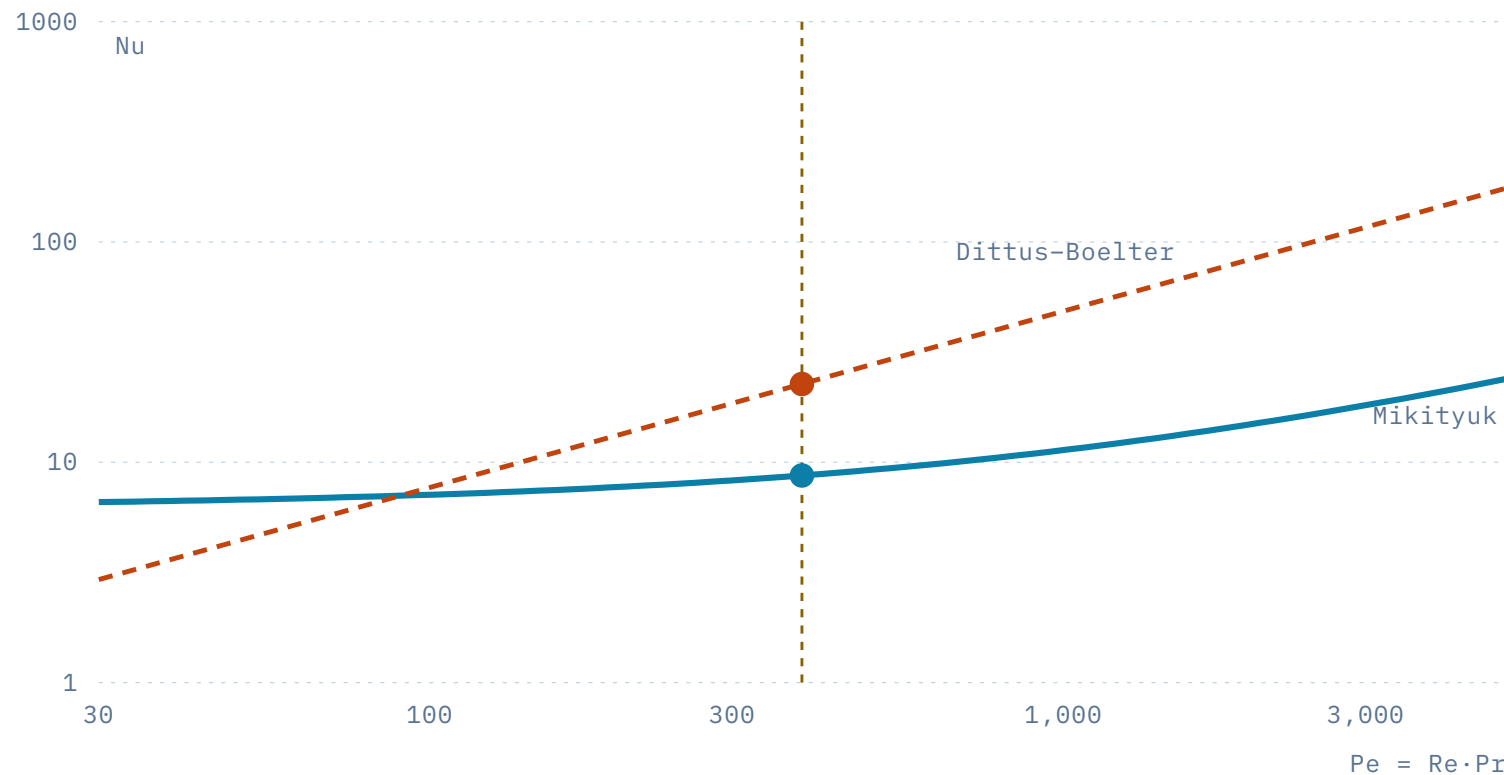
excess oxygen spoils wetting and h drops - cold traps hold it under 3 ppm

Péclet number $Pe = Re \times Pr$ - heat carried by the flow over heat conducted · **Reynolds analogy** the assumption that heat and momentum travel alike · **P/D** rod pitch over diameter

Source This textbook §7.2, Fig. 7.3 · Mikityuk (2009) · valid for $30 \leq Pe \leq 5000$, $1.1 \leq P/D \leq 1.95$ · mean error -0.1 %, RMS 1.9 % over the evaluated liquid-metal data

How wrong a **water correlation** is in a sodium rod bundle

- At a typical Pe of 387, Mikityuk gives **Nu = 8.7** ($h = 146 \text{ kW}/(\text{m}^2 \cdot \text{K})$); Dittus–Boelter gives **22.8 - 2.6 times too high**
- Inflate h 2.6-fold and the fuel reads cool, **finding margin that is not there** - exactly what happens if a code gets sodium properties but keeps its correlations



MIKITYUK (P/D 1.2)

8.7

DITTUS-BOELTER

22.7

OVERPREDICTION

×2.6

PÉCLET NUMBER PE

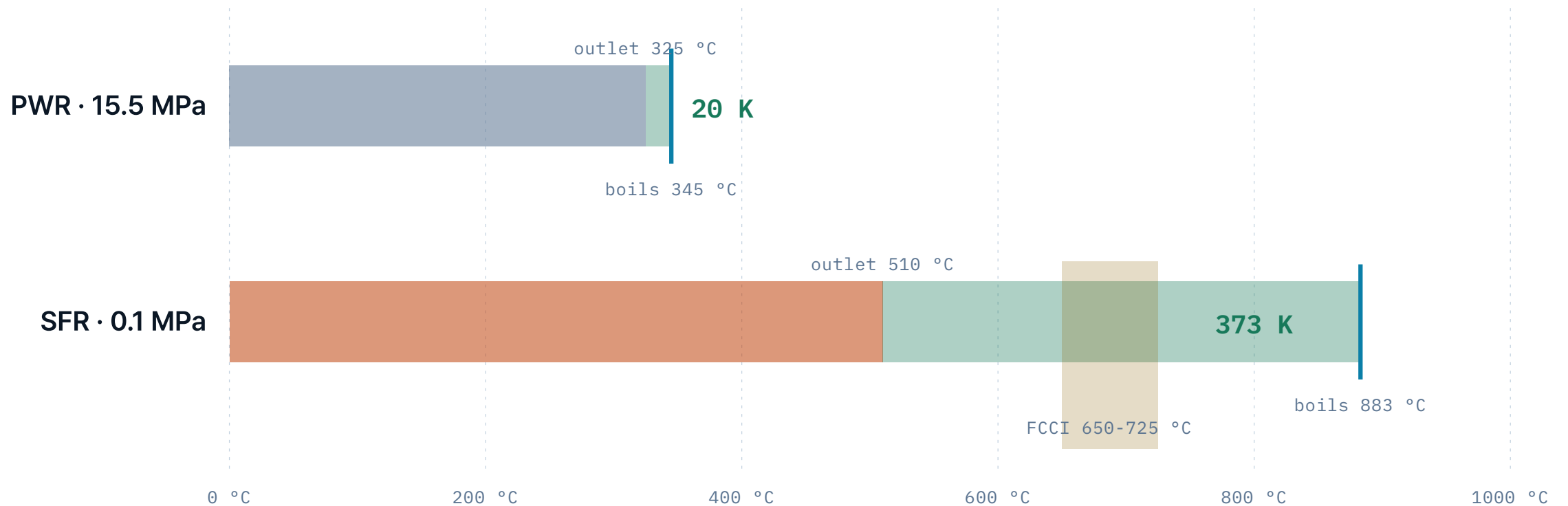
386

hydraulic diameter D_h $4 \times \text{flow area} / \text{wetted perimeter}$ · **h** heat-transfer coefficient = $\text{Nu} \cdot k / D_h$ · **rod bundle** fuel rods in a lattice

Calculation This textbook §7.3 · 450 °C, D_h 4 mm, 6 m/s, P/D 1.2 → $\text{Re} = 8.0 \times 10^4$, $\text{Pr} = 0.0049$, $\text{Pe} = 387$ · Dittus–Boelter evaluated with Pr fixed at 0.0049 and $\text{Re} = \text{Pe} / \text{Pr}$

Boiling margin - 20 K in a PWR, 330-380 K in an SFR

- All of LWR thermal hydraulics - DNBR, CHF, subchannels - is a **fight to keep those 20 K**. The SFR skips it: no pressurising (0.1 MPa argon), a thin vessel, no two-phase model, long grace time
- The ceiling is set not by the coolant but by **the metal-fuel FCCI eutectic at 650-725 °C** - hence a 510 °C outlet (ch. 8)



DNBR departure-from-nucleate-boiling ratio · **subchannel analysis** thermal hydraulics channel by channel between rods · **cover gas** argon above the free sodium surface · **FCCI** fuel-cladding chemical interaction (ch. 8)

Source This textbook §7.4 · sodium boils at 883 °C at 1 atm; outlet 500-550 °C · PWR saturation 345 °C at 15.5 MPa, outlet 325 °C · SFR margin from the 4S 510 °C outlet · 4S vessel 24 m tall, 3.5 m bore, 25 mm wall

Sodium freezing, and similarity tests - Re and Pe cannot both match

- It melts at **97.8 °C** - where water boils. Every sodium pipe carries heaters and insulation; **“safe because low pressure” always comes with “must be kept warm”**
- Water's Pr is 200 times sodium's, so **Re and Pe cannot both be matched** - flow fields come from Re·Ri similarity, heat transfer from liquid-metal tests or CFD

Re

match

inertia / viscosity

flow field, mixing

Ri

match

buoyancy / inertia

stratification, natural circulation

Pe = Re · Pr

give up

convection / conduction

Pr differs 200 ×, cannot match with Re

Where THINK Lab's RIM-PIV comes in

Sodium cannot be seen, but a transparent fluid at the same Re and Ri shows upper-plenum stratification

refractive-index-matched fluids · **distortion** the error from dimensionless groups a scaled test cannot match

Source This textbook §§7.5-7.6 · sodium melts at 97.8 °C · kept above 150 °C even when shut down · the same kind of problem as MSR freezing (459 °C) · apply the distortion assessment of *Scaling Methodology*

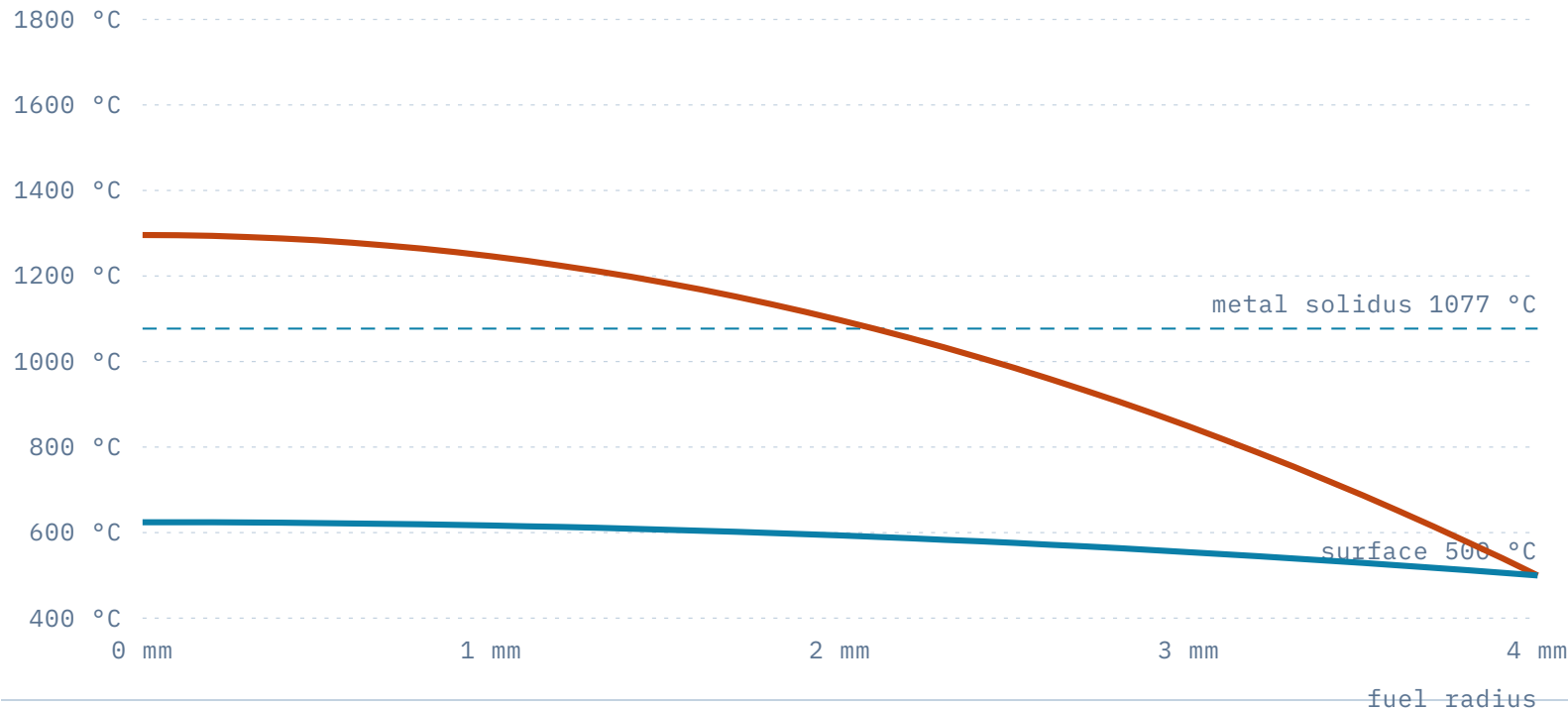
08 Metal fuel

U-Pu-Zr becomes solidus at 1077 °C; MOX at about 2700 °C. Yet the core that rode out a loss of flow without rods in 1986 was metal, and every small SFR today uses metal fuel. Thermal conductivity resolves the paradox.

Metal and oxide fuel temperatures at the same linear power

- Conductivity differs about six-fold - at 25 kW/m the centre-to-surface rise is **124 K** for metal, **796 K** for oxide
- Metal fuel runs **only 124 K** above the coolant - what makes it safe is not distance to melting but **the fuel staying tied to the coolant** (next slide)

$$\Delta T = q' / (4\pi k)$$



METAL U-10ZR CENTRE

624 °C

OXIDE MOX CENTRE

1,296 °CLINEAR POWER q' **25.0** kW/m

solidus the temperature where melting begins • **MOX** mixed uranium-plutonium oxide fuel • **linear power q'** heat per metre of fuel rod (kW/m)

Calculation This textbook §8.1, Fig. 8.1 • $\Delta T = q'/(4\pi k)$ • $k_{\text{metal}} \approx 16$, $k_{\text{oxide}} \approx 2.5$ W/(m·K) • fuel surface 500 °C, radius 4 mm (as in Fig. 8.1)

Fuel tied to the coolant - three consequences

- Metal fuel turns **the whole core into one thermometer** - safety rests on **global structural-expansion feedback** rather than local Doppler

1

Almost no stored energy

Oxide holds 796 K worth of heat at shutdown that must still come out; metal holds 124 K. Early accident behaviour is entirely different.



2

Weak Doppler feedback

Fuel temperature barely moves with power. Metal-fuel Doppler is about a third of oxide's.



3

So structural expansion dominates

Fuel follows the coolant. When flow falls, structures, sodium and rod drivelines expand together and insert more negative reactivity than Doppler.

stored energy heat held in the fuel that must still be removed after shutdown · **Doppler feedback** reactivity falling as fuel heats · **structural expansion feedback** negative reactivity from expanding core structures, sodium and rod drivelines

Source This textbook §8.1 · metal-fuel Doppler coefficient is about one third of oxide's · the experimental proof is the EBR-II tests in ch. 10

Why the smear density is 75 %

- A quarter of the space is left empty from the start - so the fuel does not press hard on the cladding before swelling reaches 30 % and the gas escapes
- The gap is filled with **liquid sodium (the bond)** - a high-conductivity bridge and a cushion until the fuel touches the cladding around 1 at.%

- 1 fission gas grows as bubbles - faster than in oxide
- 2 at about 30 % swelling the bubbles link up
- 3 gas escapes to the plenum - 70-80 % of it
- 4 leave room so the fuel does not push the cladding before then
- 5 = a 75 % smear density

burnup reached (at.%)

early



Mark-II



Mark-III



gas plenum = 1.4-1.5 × fuel volume

designs before 75 % hit their limit at 1-3 at.%

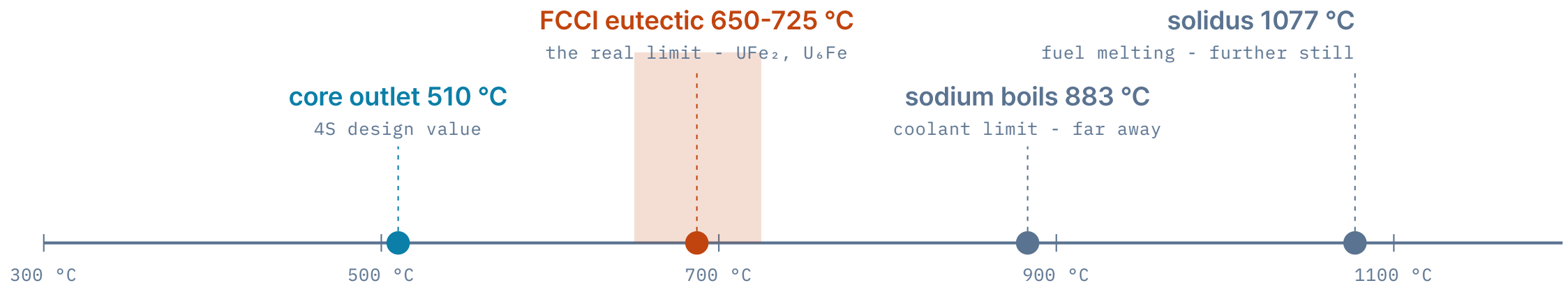
smear density fuel density if spread to the cladding bore, over theoretical density · **swelling** fuel growth from fission-gas bubbles · **at.%**

burnup in atom percent · **sodium bond** liquid sodium filling the fuel-cladding gap

Source This textbook §8.2 · early designs 1-3 at.% → EBR-II Mark-II 8-8.9 at.%, Mark-III 10 at.% · gas plenum 1.4-1.5 times the fuel volume

What sets the temperature limit - the FCCI eutectic, not melting

- Coolant is liquid to 883 °C and fuel solid to 1077 °C, yet **the real limit is a metallurgical eutectic at 650-725 °C** - hence a 510 °C outlet
- **Knowing only "it boils at 883 °C, so lots of margin" is half the story** - the design is set by 650 °C. Miss that in a talk and you will be corrected



Two things happen in FCCI

- ① lanthanides and Pd erode the cladding bore (up to 170 μm)
- ② low-melting U-Fe and Pu-Fe eutectics - the worse of the two

FCCI fuel-cladding chemical interaction · **eutectic** a mixture melting below either component · **lanthanides** the rare-earth fission products, $Z = 57-71$ · **HT-9** a ferritic-martensitic cladding steel

Source This textbook §8.3 · lanthanides (Ce, Pr, Nd, La, Sm) and Pd collect at the interface and erode the cladding - up to 170 μm measured (HT-9, U-23Zr, 660 °C) · U-Fe and Pu-Fe eutectics at 650-725 °C

Metal-fuel **cladding** and demonstrated burnup, and the 4S fuel pin

- HT-9 became standard because ferritic-martensitic steel **resists void swelling** under fast neutrons. 4S, Natrium and PRISM all use it
- **Half of the 4S pin is empty gas plenum** - thirty years without refuelling means that much fission gas to hold

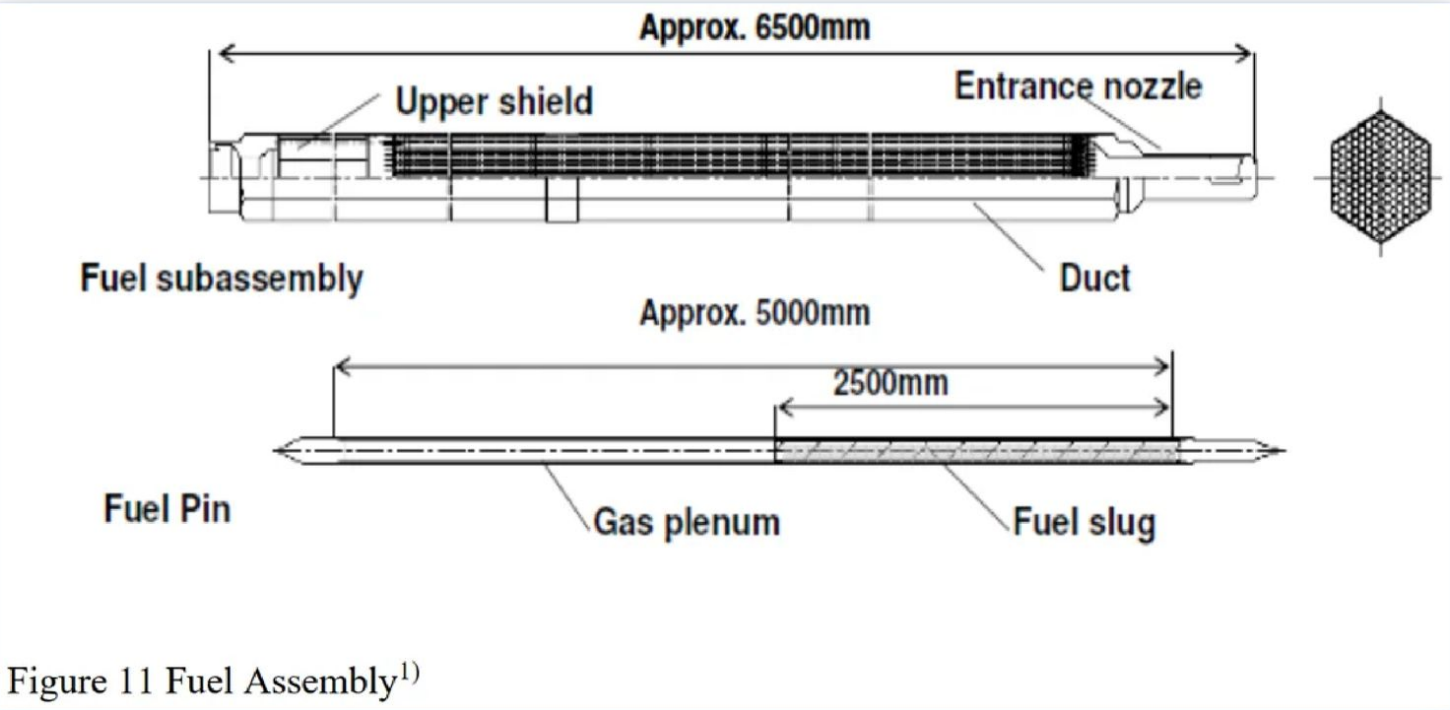


Figure 11 Fuel Assembly¹⁾

CLADDING	TYPE	BURNUP	NOTE
316 SS	austenitic	8~8.9 at. %	EBR-II Mark-II
D9	mod. austenitic	19.8 at. %	FFTF
HT-9	ferritic-martensitic	19.9 at. %	FFTF · 643 °C peak
ODS	oxide-dispersion	—	not qualified

4S assembly (about 6500 mm) and pin (about 5000 mm) - 2500 mm of fuel slug, the other half gas plenum · textbook Fig. 8.2

void swelling growth of metal under fast-neutron irradiation as voids form · **ODS** oxide-dispersion-strengthened steel · **FFTF** the US Fast Flux Test Facility · **slumping** soft fuel sagging under its own weight

Source This textbook §8.4, Table 8.1, Fig. 8.2 · full-length 91.4 cm pins reached 19.9 at.% in FFTF without slumping · 4S assembly 6500 mm; of the 5000 mm pin, 2500 mm is fuel slug

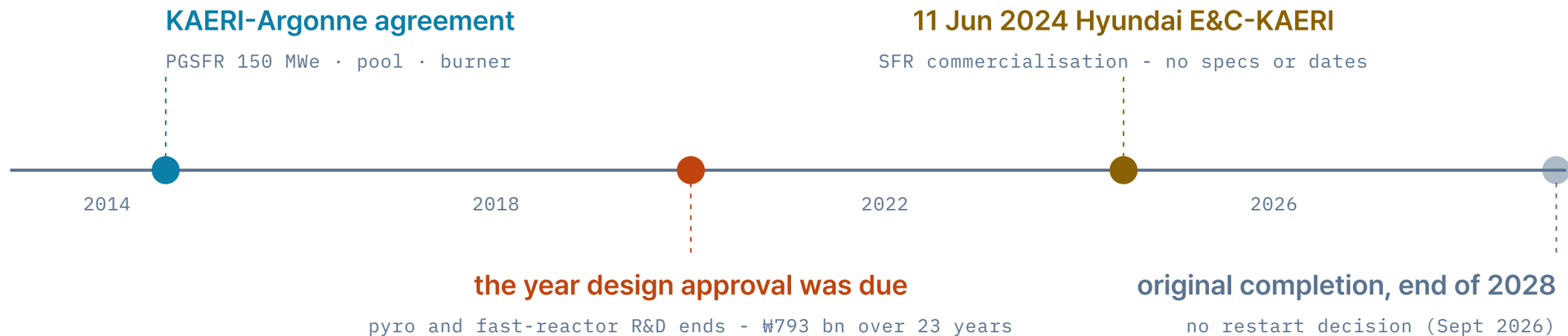
Metal against oxide fuel - a summary table

- Do not read oxide as inferior - it has **far more fabrication and reprocessing experience**, is stable hot, and uses LWR fuel infrastructure. Metal wins in **small, passive-safety designs**

ITEM	METAL (U-ZR, U-Pu-ZR)	OXIDE (MOX)
Conductivity	about 16 W/(m·K)	2~3 W/(m·K)
Solidus	1077 °C	about 2700 °C
Smear density	75 % (design)	85~95 %
Fuel-coolant ΔT	small	large
Doppler feedback	weak	strong
Stored energy	little	much
Overpower margin (TOP)	4-4.5 × rated	2.5-3 ×
Heavy-metal density	high → harder spectrum, better breeding and burning	lower
Real temperature limit	FeCr eutectic 650-725 °C	melting

Pyroprocessing, and what became of Korea's PGSFR

- Metal fuel and pyroprocessing go together - U and the transuranics come out **together as metal** and are cast directly. No pure-plutonium step is the **proliferation-resistance argument**
- **"The research is alive; there is no demonstration project"** is exactly where Korean SFR stands in 2026



pyroprocessing dry electrochemical treatment of spent fuel in molten salt · **electroreduction, electrorefining** turning oxide into metal, then separating metals · **PGSFR** Korea's Prototype Gen-IV SFR (150 MWe)

Source This textbook §8.6 · about ₩793 bn over 23 years before pyroprocessing and fast-reactor R&D ended in 2020 · **as of September 2026** no decision to restart

09 Taking 4S apart

What does a reactor that is not refuelled for thirty years look like? 4S is useful for teaching because its design is radically simple. You can see clearly how one goal - thirty years without refuelling - reshapes every component.

Pool and loop - why every modern small SFR is a pool

- Passive decay-heat removal (RVACS) works through **the vessel surface**, so the core must be inside it, and **a large sodium inventory** buys the first hours

	POOL	LOOP
Primary sodium	large (EBR-II about 333 m ³)	small
Thermal inertia	very large → slow transients	small
Primary leaks	caught by the guard vessel - no loss-of-sodium path	pipes are a leak path · double wall
Natural circulation	easy (height available)	harder
Drawback	large vessel, awkward inspection and refuelling	long sodium piping, more fire exposure

Why **natural circulation** works so well in an SFR

- One formula holds every reason - **ΔH is large, ΔT can be large, ρ is large, and viscosity is low**
- So an SFR can **remove decay heat by natural circulation alone** once forced flow stops - measured at EBR-II and Phénix

$$\dot{m} \propto \sqrt{(2\rho^2 A^2 g \beta \Delta T \Delta H / K)}$$

ΔH height

20 m-class pool vessel, core-IHX a few m

ΔT temperature rise

330 K to boiling - 100 K in an accident is fine

ρ density

850 kg/m³ - near water, 260 × helium

ν kinematic viscosity

3.2×10⁻⁷ m²/s - higher Re, less friction

P · AGAINST HELIUM

×260

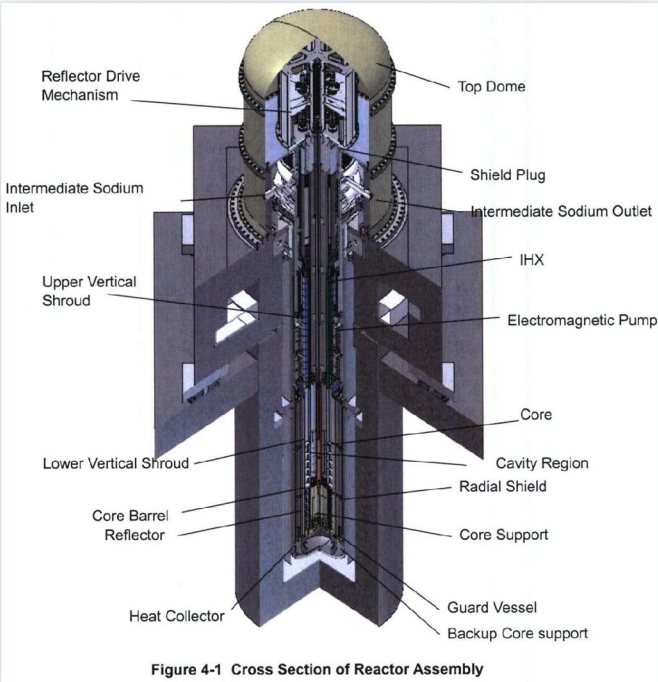
VISCOSITY · VS WATER AT 20 °C

1/3

β thermal expansion coefficient · ΔH height between heat source (core) and sink (HX) · K total loop loss coefficient · **kinematic viscosity**
viscosity over density

Source This textbook §9.2 · buoyancy $\rho g \beta \Delta T \Delta H$ balancing friction $\frac{1}{2} K \dot{m}^2 / (\rho A^2)$ · sodium kinematic viscosity at 400 °C 3.2×10⁻⁷ m²/s · the EBR-II and Phénix natural-circulation tests are today's validation standard

Toshiba 4S - principal design data



4S reactor assembly section · Fig. 9.1

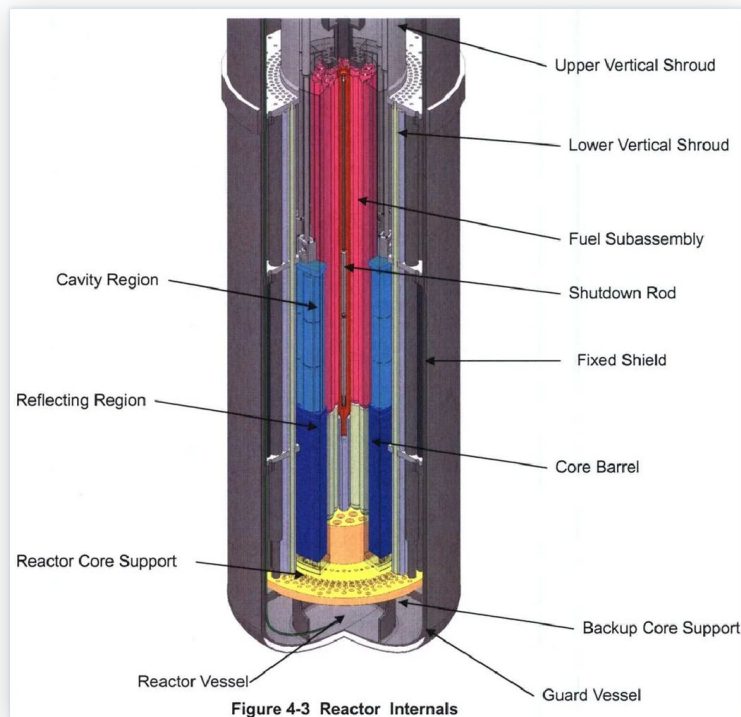
ITEM	VALUE
Output · type	30 MWt / 10 MWe · pool
Core life	30 years, no refuelling
Primary sodium	355 → 510 °C
Reactivity control	movable annular reflector - 1-1.5 mm a week
Shutdown	shutdown rod dropped by gravity
Pumps	two annular linear-induction EM pumps in series · 10.6 m³/min · 0.05 MPa each
Fuel	sodium-bonded U-10Zr · 17 % / 19 % · HT-9 · 34,000 MWd/t
Reactor vessel	24 m tall · 3.5 m bore · 25 mm wall · 304 SS
Intermediate loop	482 t/h · 310 → 485 °C · double-wall helical SG
Steam	10.5 MPa / 453 °C · 44.2 t/h

4S Super Safe, Small and Simple · **U-10Zr** uranium alloy with 10 % zirconium · **MWd/t** energy per tonne of fuel (burnup) · **304 SS** an austenitic stainless steel

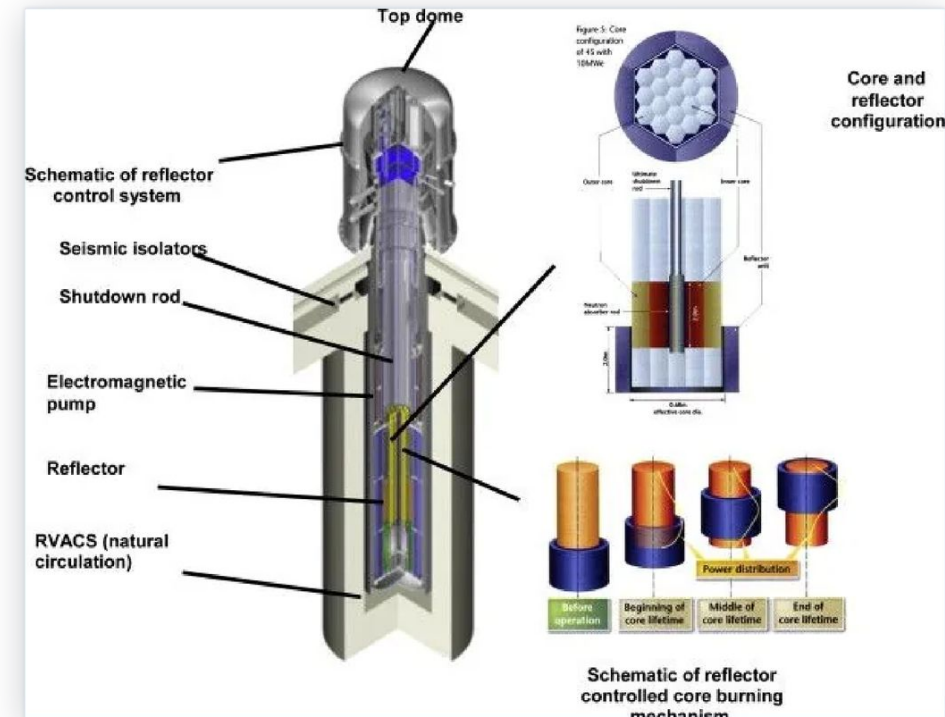
Source This textbook §9.3, Table 9.2, Fig. 9.1 · NRC pre-application for Galena, Alaska stopped in 2013 · Zap Energy announced a 4S revival in April 2026 (CEO named 29 April) · no licensing or deployment schedule

4S internals and the full cutaway

- Top to bottom: reflector drive, shield plug, intermediate sodium nozzles, IHX, EM pumps, core, guard vessel - **everything stacked in one vessel**
- Core support is **doubled** (support structure plus backup), and the reactor sits on seismic isolators



4S reactor internals · Fig. 9.2



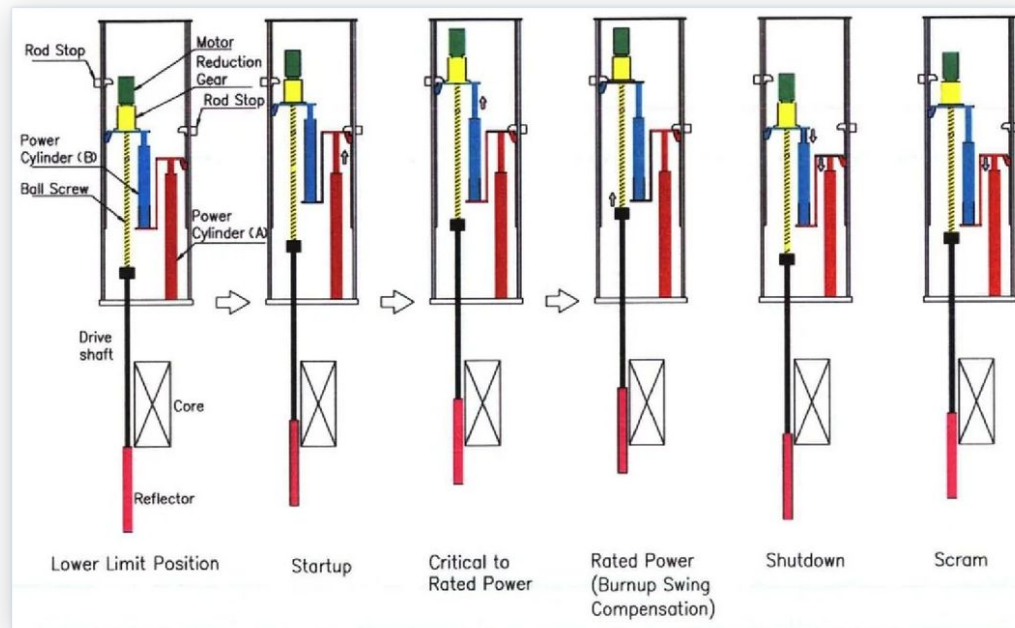
4S cutaway and the reflector-control principle · Fig. 9.5

shroud a cylindrical partition dividing flow paths · **seismic isolator** bearings that keep ground motion out of the plant · **RVACS** Reactor Vessel Auxiliary Cooling System

Source This textbook §9.3, Figs. 9.2 and 9.5 · upper and lower vertical shrouds split the flow paths; core support and backup core support are doubled

A reflector instead of rods - reactivity control in 4S

- The core stays critical only as high as the reflector covers - as fuel burns, the reflector rises. **1-1.5 mm a week for thirty years sweeps the whole core height**
- Rod withdrawal runs at tens of cm a second; the reflector drive is **mechanically millimetres a week** - **the maximum insertion rate is physically small**



Reflector sequence - lower limit → start-up → critical and rated → burnup swing → shutdown → scram
· Fig. 9.3

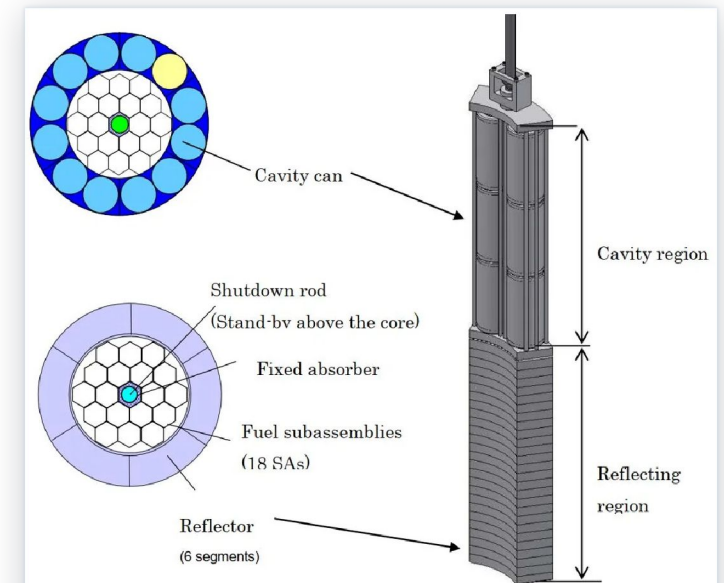


Figure 13 Core Cross Sectional View and Reflector Configuration

Core section and reflector layout · Fig. 9.4

reflector a structure that returns escaping neutrons · **burnup compensation** making up reactivity lost as fuel burns · **reactivity-insertion**

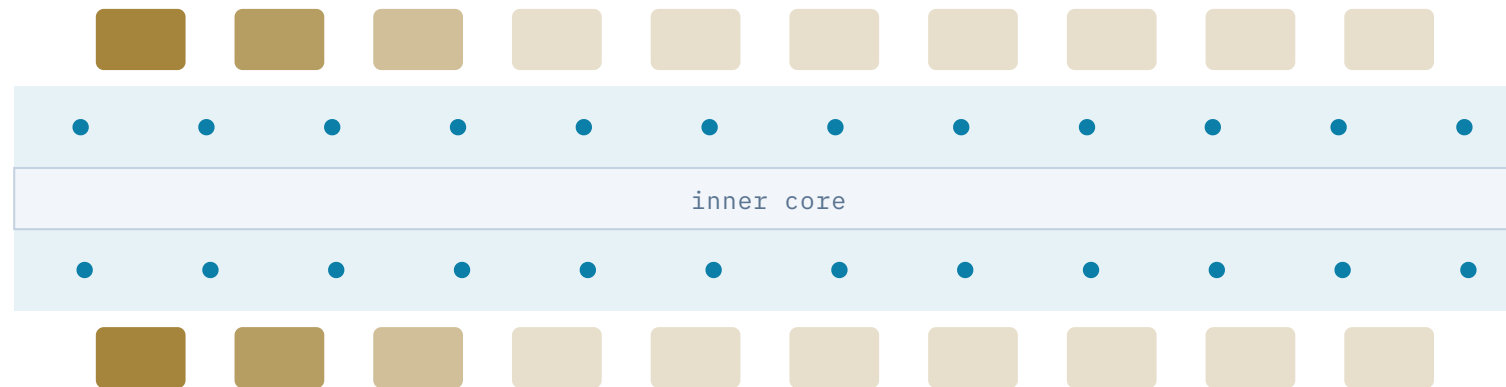
accident a sudden reactivity jump and power surge · **scram** emergency shutdown

Source This textbook §9.3, Figs. 9.3-9.4 · 18 fuel assemblies, a fixed central absorber, a shutdown rod waiting above the core, a six-segment reflector

The **electromagnetic pump** - $F = J \times B$, and no inertia

- Sodium conducts electricity - no rotating parts, shaft seals or bearings. Low head suffices **because a pool has almost no piping resistance**
- Weaknesses - **under 40 % efficient**, and **no inertia**: flow drops the instant power goes. A rotating pump coasts for tens of seconds on a flywheel - ch. 10 shows why that matters

coils - travelling field B



annular sodium channel - induced J, force F

flow →

$$F = J \times B$$

4S · TWO IN SERIES

10.6 m³/min

HEAD (EACH)

0.05 MPa

INERTIA FIX

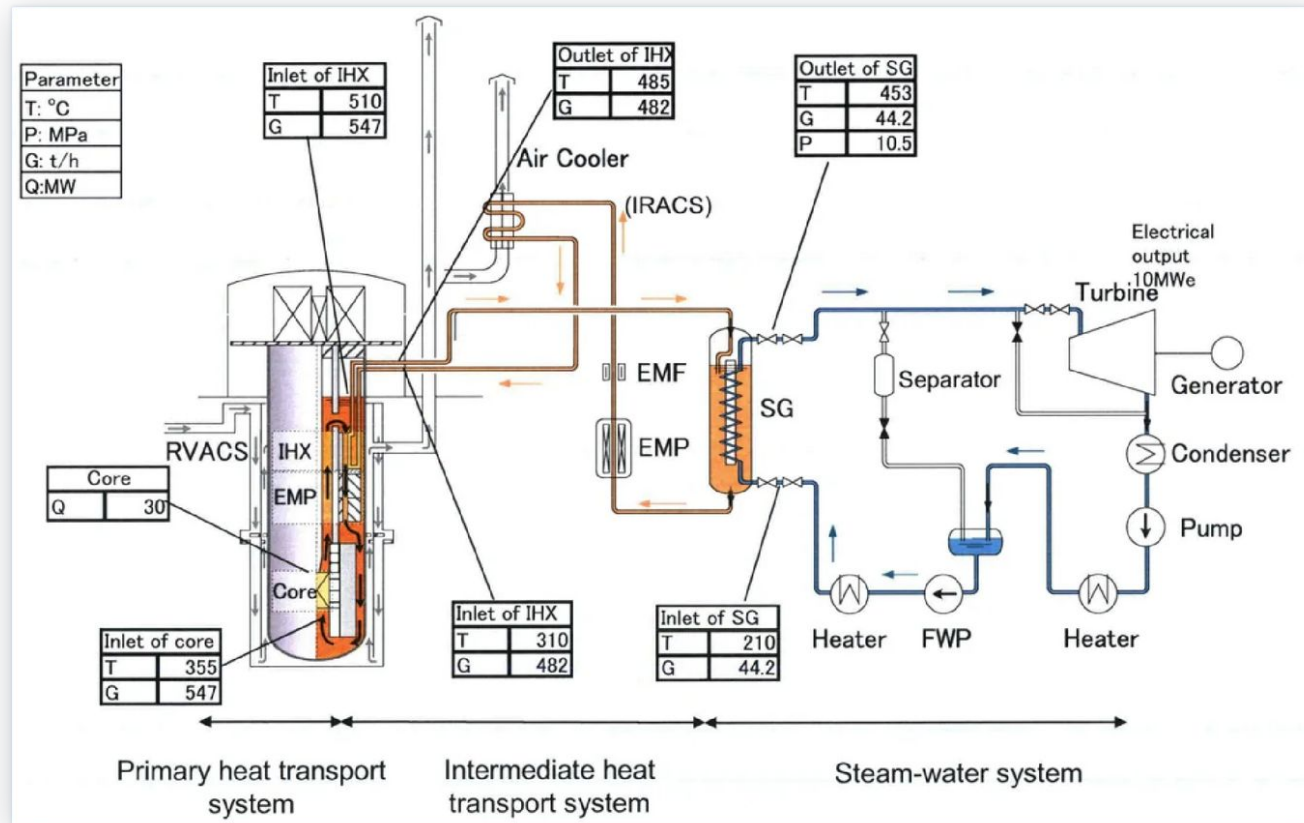
**a synchronous generator or
battery circuit mimics inertia**

Lorentz force the force on a current in a magnetic field · **linear induction** a travelling field from coils induces current in the fluid and pushes it · **coast-down** flow decaying after pump power is lost · **flywheel** a heavy wheel adding rotational inertia

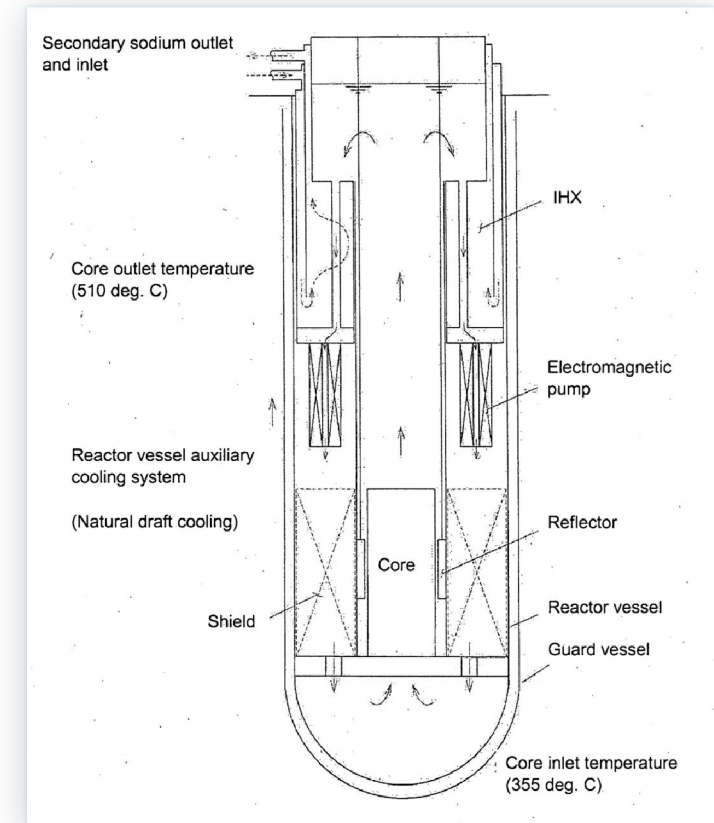
Source This textbook §9.3 · 4S uses two annular linear-induction pumps in series, 10.6 m³/min in all, 0.05 MPa each · efficiency usually under 40 % · the drawing is a principle sketch

4S heat transport - primary, intermediate, steam

- Primary **355/510 °C** → intermediate **310/485 °C** → steam **453 °C**. The temperature lost at each step is the price of the intermediate loop (end of ch. 9)



4S heat transport - primary, intermediate, steam-water · Fig. 9.6



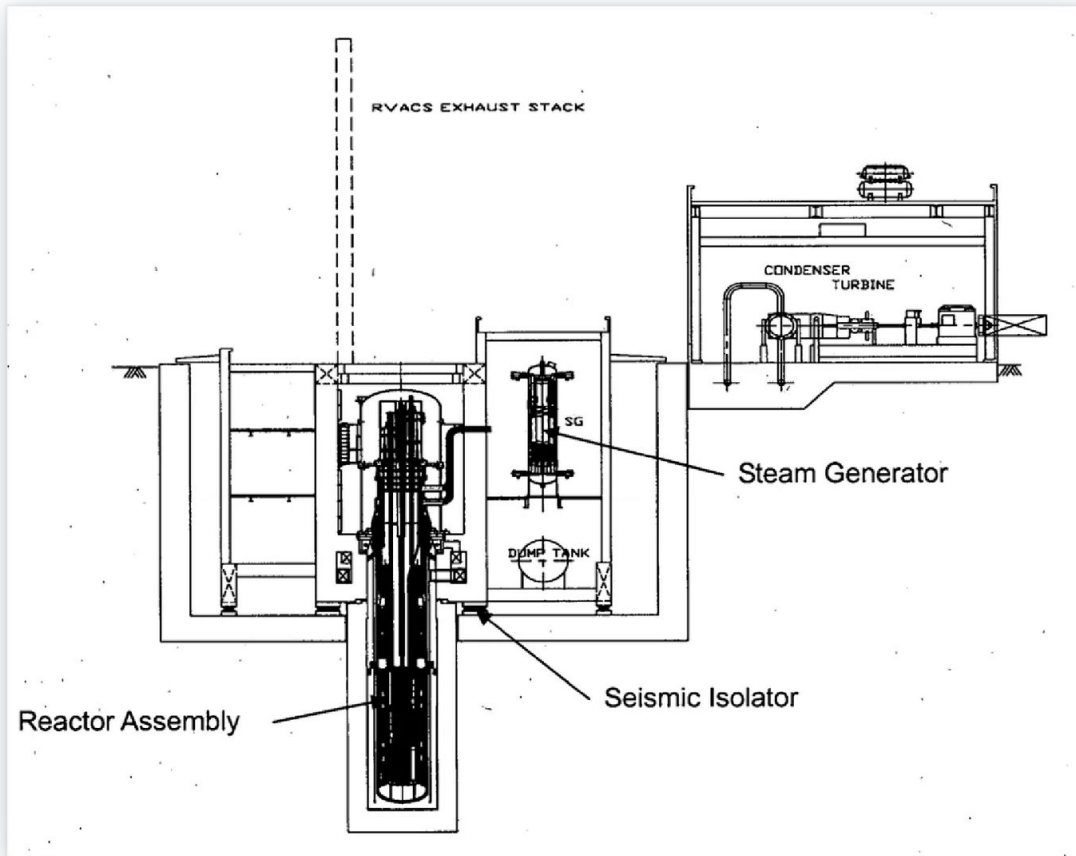
4S system concept · Fig. 9.7

IRACS an air cooler on the intermediate loop · **EMP, EMF** electromagnetic pump and flowmeter · **FWP** feedwater pump

Source This textbook §9.3, Figs. 9.6-9.7 · primary 355/510 °C → intermediate 310/485 °C → steam at 453 °C · 10 MWe · RVACS is natural draught outside the guard vessel

4S plant layout and fuel handling

- The reactor sits **below grade on seismic isolators**; only the RVACS stack comes up
- The core is **replaced whole once every thirty years** - fuel handling becomes swap-out equipment rather than a standing system



4S plant layout · Fig. 9.8

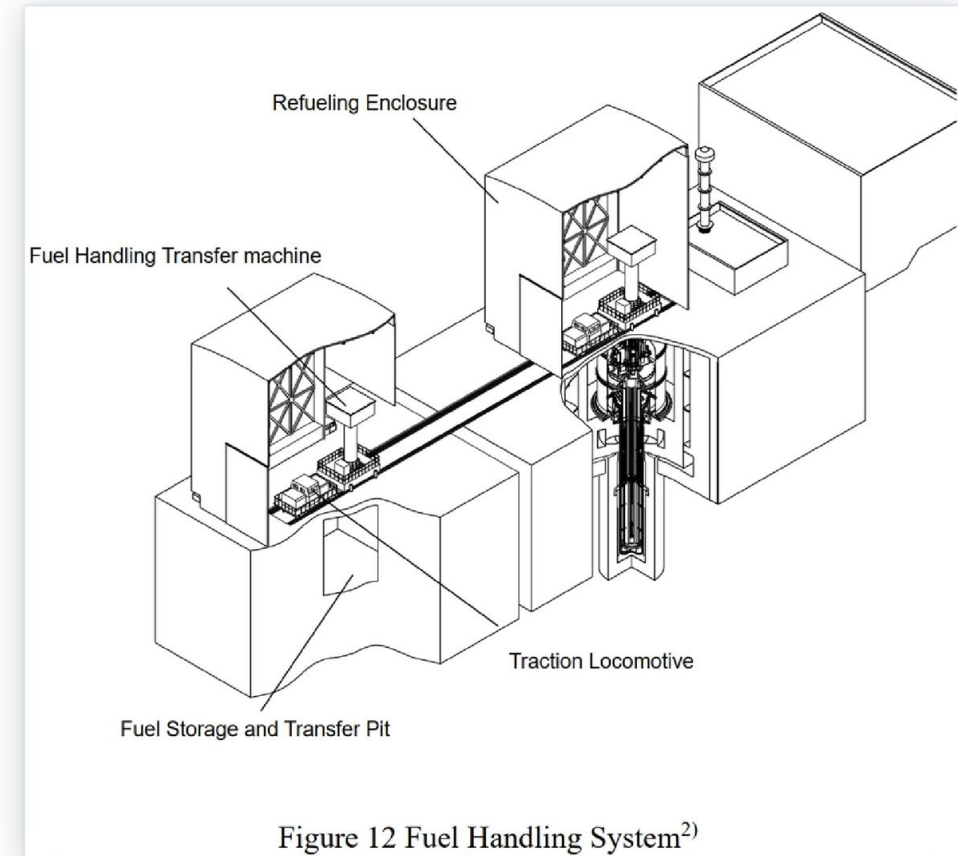


Figure 12 Fuel Handling System²⁾

Fuel handling system · Fig. 9.9

Russia's BN series - the only long operating record

- BN-600 has run **since 8 April 1980** - the longest-running power fast reactor, and **the strongest evidence that SFRs actually work**



BN-600 core section model · Fig. 9.10



BN-800 reactor top and sodium piping · Fig. 9.11

BN-600 · 1470 MWT / 600 MWE

**76 % over 30 years · 114 TWh ·
burnup 7 → 11.4 % · licensed
to 2040**

BN-800 · 2100 MWT / 864 MWE

**commercial Oct 2016 · over 60
% MOX, heading for a full-MOX
core**

BN-1200M · 1220 MWE

**planned as unit 5 · around
2035**

BN Russian for “fast neutron” · **capacity factor** actual output over output at rated power all the time · **TWh** a billion kWh · **full-MOX core** a core fuelled entirely with MOX

Source This textbook §9.4, Figs. 9.10-9.11 · as of September 2026 · Beloyarsk units 3 and 4

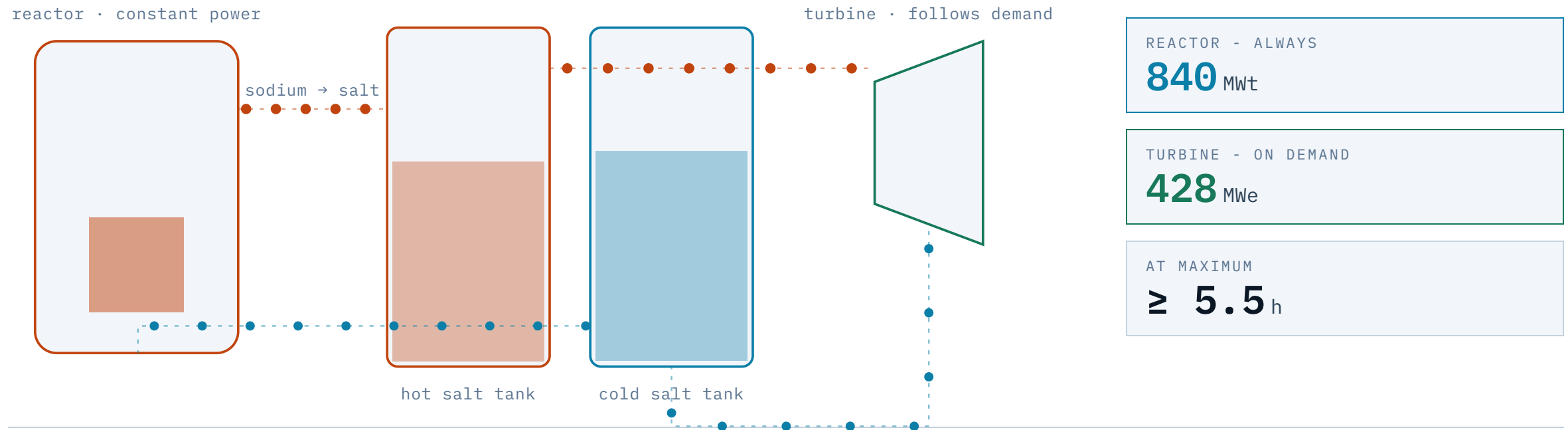
SFR status in the US, Canada, India and China

- The Natrium construction permit (March 2026) is the first commercial-reactor permit in about ten years and the first non-LWR one in forty
- Oklo’s 2020 heat-pipe Aurora was dropped - today it is a 75 MWe sodium pool. Do not quote the old material

DESIGN	SIZE · TYPE	STATUS (SEPT 2026)
TerraPower Natrium (US)	840 MWt / 345 MWe (500 max)	non-nuclear works June 2024 → NRC permit March 2026 → nuclear construction 24 April 2026 (Kemmerer, WY). Completion target 2030; up to eight units for Meta
ARC-100 (Canada)	286 MWt / 100 MWe · pool · metal	EBR-II lineage. Site-preparation application June 2023, CNSC VDR done March 2025. No construction licence
Oklo Aurora (US)	75 MWe · pool · metal HALEU	NRC rejected in Jan 2022 → DOE pilot route. Ground broken 22 Sept 2025; preliminary safety analysis approved 11 June 2026 (3rd of 4 steps); operation target 2028
PFBR (India)	1250 MWt / 500 MWe · MOX	construction Oct 2004, first criticality 8 April 2026 - 15+ years late. No grid date
CFR-600 (China)	600 MWe gross	unit 1 started Dec 2017, running since 2023. Unit 2 started Dec 2020. Thermal power quoted as 1500 or 1882 MWt

Natrium - molten-salt storage between reactor and turbine

- The reactor runs **at constant power** and the storage pushes the turbine up and down with demand - the old nuclear weakness, load following, **sidestepped with heat storage**
- Valuable on grids heavy with renewables - 345 MWe rated, **500 MWe for over 5.5 hours**, 10 % a minute



nitrate salt the storage medium from solar thermal plants ($\text{NaNO}_3\text{--KNO}_3$) · **load following** raising and lowering output with demand · **storage tanks** separate hot and cold salt tanks

Source This textbook §9.4 · 840 MWt / 345 MWe rated, 500 MWe max · over 5.5 h at maximum · 10 % a minute ramp · the drawing is a concept sketch

For reference - **lead-cooled reactors** and the trade against sodium

- Sodium **trades chemical reactivity for good flow and materials compatibility** - lead makes the opposite trade

What lead gains

- no reaction with water → no intermediate loop
- does not burn
- heavy, barely moderates

What lead gives up

- melts at 327 °C → a real freezing risk
- corrodes and erodes steel
- dense → heavier structural and seismic loads

SEALER / Blykalla

Sweden · 80 MWt demonstrator · six commercial units, 330 MWe · construction application 18 May 2026 · early 2030s

BREST-OD-300

Russia · 300 MWe · nitride fuel · physics start-up 2026, grid early 2027

LFR lead-cooled fast reactor · **erosion** a flowing fluid wearing a surface away · **nitride fuel** uranium-plutonium nitride - dense and conductive

Source This textbook §9.4 · **as of September 2026** · SEALER-D 80 MWt; commercial plan six units, 330 MWe at Gävle-Norrundet; construction application 18 May 2026 · BREST-OD-300 uses nitride fuel; physics start-up 2026, grid in first half of 2027

Reading an NEA SMR Dashboard **design card**

- The six gauges measure **project maturity**, not design specifications
- **This edition is out of date** - numbers like Natrium's 840 MWth / 500 °C still hold, but the gauges have moved several notches. Always check the live version



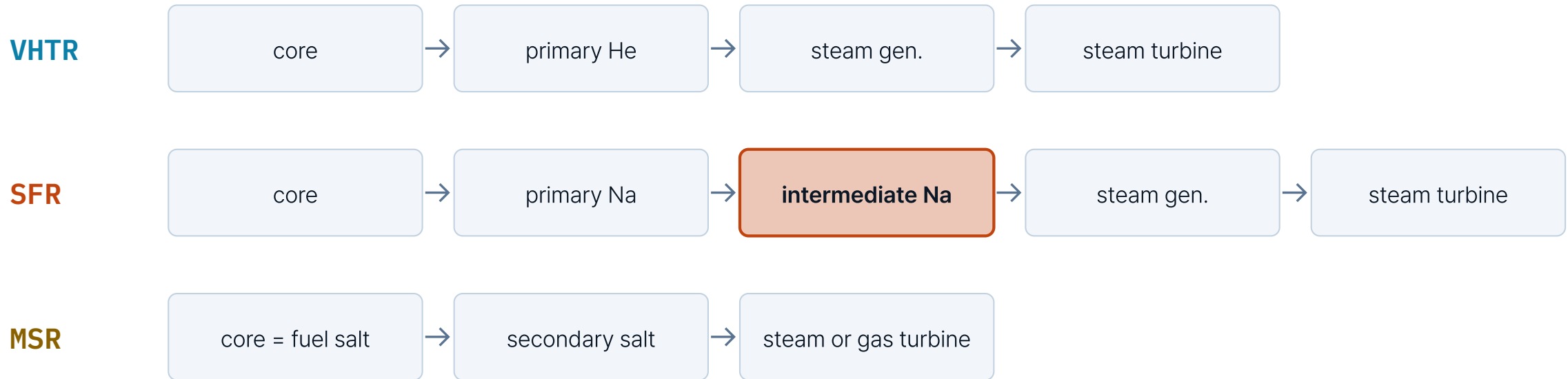
NEA SMR Dashboard (2023) design card - Natrium · Fig. 9.12

NEA the OECD Nuclear Energy Agency · **project maturity** progress on six axes - licensing, siting, financing, supply chain, engagement, fuel

Source This textbook §9.5, Fig. 9.12 · NEA SMR Dashboard, **2023 edition** - now a digital dashboard with different fields and values · see *Finding SMR Data* §2.4

Heat paths by reactor type, and what the SFR **intermediate loop** is worth

- Two reasons - ① **primary sodium is activated** (Na-24, 15 h) and cannot go to the turbine hall · ② **a failed SG tube never brings water to the core**
- The real sodium-water danger is **propagation**, not explosion. Hardest is the **intermediate leak** - too small for the rupture disc, too fast for micro-leak detection. Why 4S uses double-wall tubes



only the SFR adds a loop - to keep sodium and water apart

wastage a reaction jet eating into neighbouring tubes · **overheating** loss of tube strength over a wide area · **rupture disc** a thin plate that bursts to relieve pressure · **double-wall tube** two walls with leak detection between

Source This textbook §9.6, Fig. 9.13 · Na-23 absorbing a neutron becomes Na-24 (15 h half-life), a strong gamma emitter · sodium-water reactions propagate by wastage and overheating

10 The reactor that stops itself

On 3 April 1986 EBR-II lost cooling twice at full power and both times shut itself down with no rods and no operator. Every small SFR's claim to "passive safety" is a claim to reproduce what was measured that day.

Two faces of SFR safety - five good, three bad

- The five good points together make **a design that survives accidents without rods**
- The last bad point is the worst - an LWR loses reactivity when coolant leaves, but **an SFR can go the other way**. Handling it is half of core design

Good news

- 1 low pressure - no rapid depressurisation
- 2 330 K boiling margin
- 3 a huge pool thermal inertia
- 4 strong natural circulation
- 5 metal fuel: structural-expansion feedback

Bad news

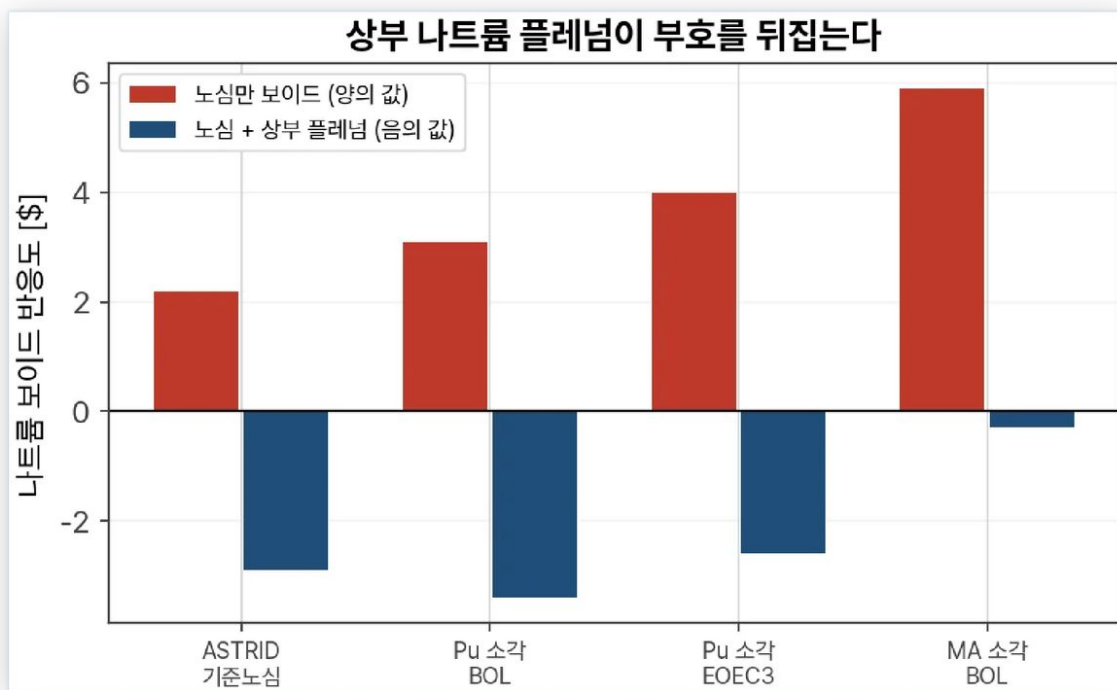
- 1 sodium explodes with water
- 2 sodium burns in air
- 3 **a large core has positive void reactivity**

void reactivity the reactivity change as coolant leaves · **rapid depressurisation** a high-pressure system losing pressure at once through a break

Source This textbook §10.1 · in an LWR voids remove moderation and reactivity falls - the foundation of LWR safety

The two components of sodium void reactivity, and ASTRID core values

- ① **Spectral hardening (positive)** scales with volume, ② **extra leakage (negative)** with surface-to-volume - large cores go positive, small flat ones can go negative
- Void the core alone and it is positive; void **the upper sodium plenum too** and it flips negative



The upper sodium plenum flips the sign - red core only, blue core plus upper plenum · textbook
Fig. 10.1

$$\Delta\rho_{\text{void}} = \Delta\rho_{\text{spectrum}} + \Delta\rho_{\text{leakage}}$$

CORE	CORE	+ PLENUM	DOPPLER	BEFF
ASTRID ref.	+2.2 \$	-2.9 \$	-870	—
Pu burner BOL	+3.1 \$	-3.4 \$	-571	331
Pu burner EOC3	+4.0 \$	-2.6 \$	-540	328
MA burner BOL	+5.9 \$	-0.3 \$	-275	275

Doppler constant and β_{eff} in pcm

spectral hardening neutron energy rising - Pu-239's η goes up · **leakage** neutrons escaping the core · **β_{eff}** effective delayed-neutron fraction
· \$ reactivity divided by β_{eff}

Source This textbook §10.2, Table 10.1, Fig. 10.1 · the French ASTRID family of cores · BOL beginning of life · EOC3 end of equilibrium cycle 3 · MA minor actinides

Three features of the low-void (CFV) core, and the price of burning MA

- Loading minor actinides to burn them **worsens all three indicators at once** - waste burning and safety margin collide head-on.
Nothing in design is free

1 A large upper sodium plenum, absorber above

95 cm inner / 76 cm outer. When it voids, neutrons pour out upward

2 An axially heterogeneous, short core

effective height cut about 20 % (50/70 cm) - more leakage

3 No breeding blanket

this works against you - 1 and 2 must compensate

CORE-ONLY VOID WORTH

+2.2 → +5.9 \$

DOPPLER CONSTANT

-870 → -275 pcm

BEFF

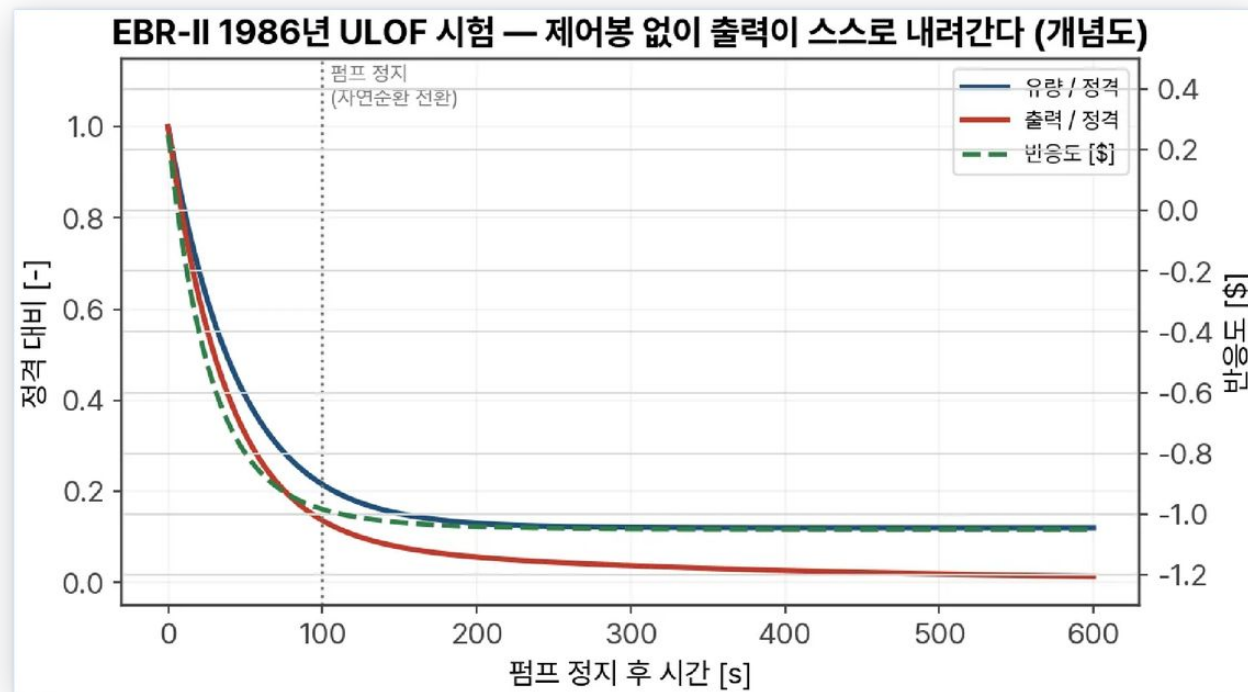
331 → 275 pcm

CFV Cœur à Faible effet de Vide - the low-void core · MA minor actinides (Np, Am, Cm) · **breeding blanket** U-238 round the core · **heterogeneous** alternating fuel and non-fuel layers

Source This textbook §10.2 · upper sodium plenum 95 cm inner / 76 cm outer · effective height cut about 20 % (50/70 cm) · last row of Table 10.1

EBR-II, 3 April 1986, test 1 - loss of flow without scram

- Station blackout simulated - the low-flow trip defeated and the main pumps cut. **They stopped in about 100 s**, leaving natural convection and a battery-powered auxiliary pump
- As the coolant heated, negative reactivity came in and **power fell to practically zero** - no rods, no operator



Conceptual behaviour in the EBR-II ULOF test - flow falls, temperature rises, negative reactivity follows in proportion and power drops · textbook Fig. 10.2

WHERE THE NEGATIVE REACTIVITY CAME FROM

thermal expansion of core structure, sodium and rod drivelines

FUEL SIDE (DOPPLER + EXPANSION)
the smaller term

MAGNITUDE

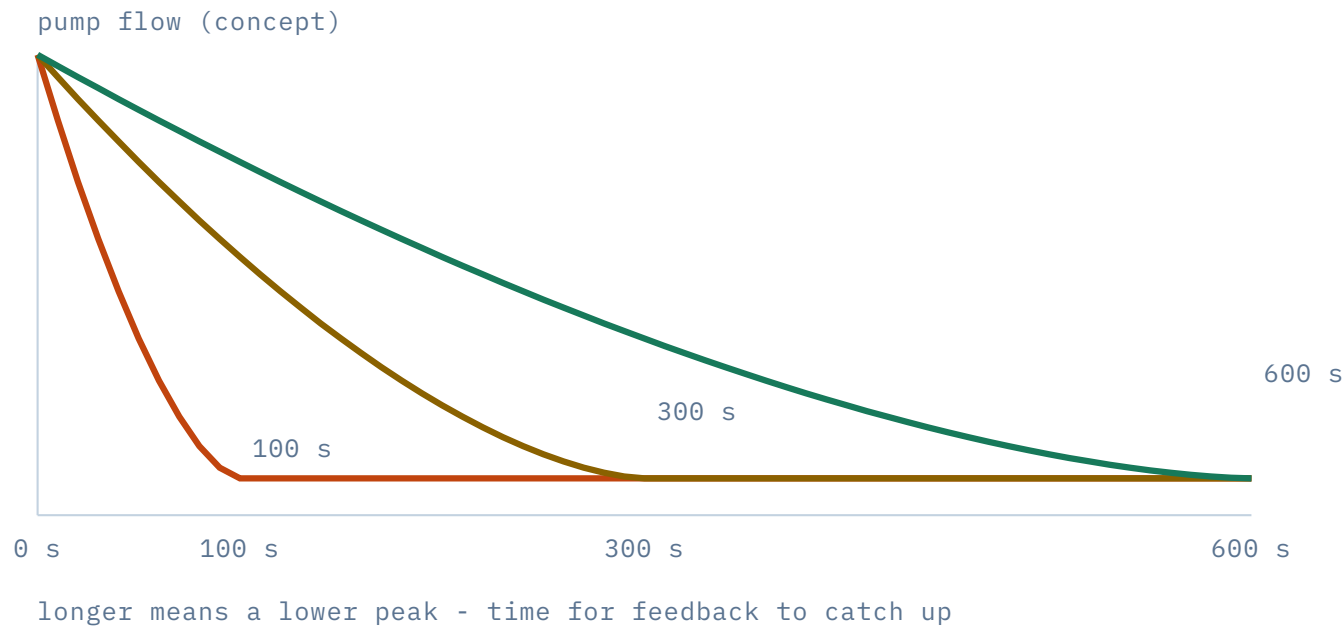
roughly proportional to core ΔT

LOFWS Loss of Flow Without Scram · **SBO** station blackout · **coast-down** flow decaying after pump power is lost · **ULOF** unprotected loss of flow

Source This textbook §10.3, Fig. 10.2 · EBR-II 62.5 MWt / 20 MWe pool · about 333 m³ of primary sodium · pool 371 °C · mixed outlet 473 °C · U alloy (66 %) · 316 SS · 91 pins per assembly · 34 cm active core · full power · over 60 observers (13 foreign)

Pump coast-down time is a design parameter

- The longer the coast-down, the lower the peak - flow must fall **slower than the neutron and thermal time constants** so feedback can keep up
- Designers tune pump and motor **inertia** to get the coast-down they want - which is why ch. 9 flagged that **EM pumps have none**



β differs about 2 × between U-235 and Pu-239

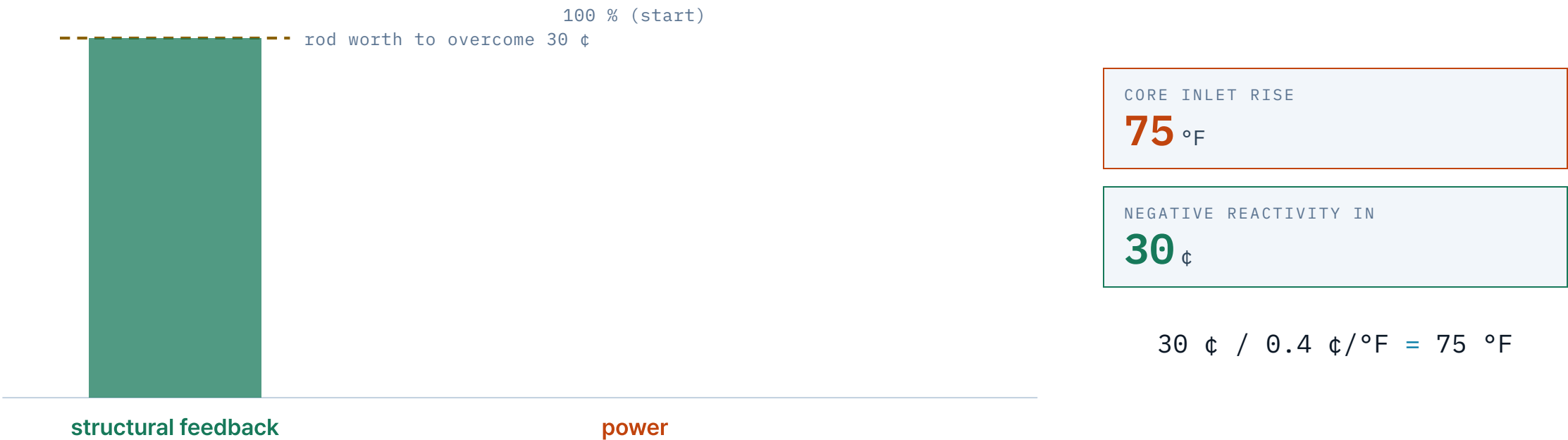
As the core moves from uranium to plutonium equilibrium the dollar scale changes wholesale. Set the coast-down for the plutonium equilibrium core - set it for the first core and the margin vanishes late in life.

delayed neutrons neutrons emitted seconds after fission that make a reactor controllable · **time constant** the characteristic response time · **CRBRP**, **FFTF** US fast-reactor project and test facility

Source This textbook §10.3 · repeated with 100, 300 and 600 s coast-downs · fuel-cladding interface limit for anticipated events 1319 °F (715 °C) · only tests 44 and 45 were predicted to exceed it briefly in some assemblies · zero fuel failures - the fuel went on to normal burnup · the flow curves are concept shapes only (the book gives no curve data)

EBR-II test 2 - loss of heat sink without scram

- Back at full power, **the secondary pumps were tripped** - no heat could leave. As the core inlet warmed, the lower reflector and core support expanded and inserted negative reactivity
- The reactor shut down after the inlet rose **about 75 °F (42 °C)** - $30 \text{ ¢} \div 0.4 \text{ ¢/°F}$. These two tests are **the template for every small-SFR safety case** today

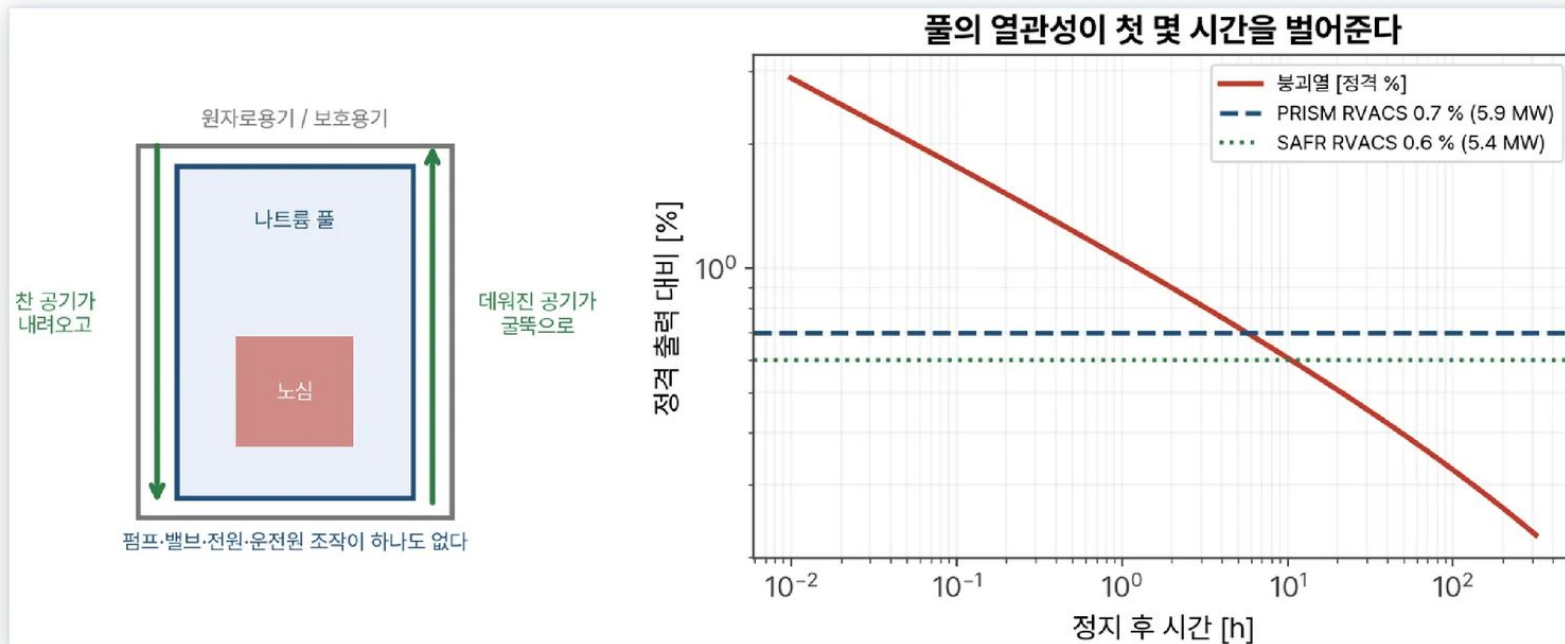


LOHSWS Loss of Heat Sink Without Scram · ¢ (cent) 1/100 of a dollar of reactivity · inlet temperature coefficient reactivity per degree of core-inlet temperature

Calculation This textbook §10.3 · rod worth to overcome about 30 ¢ ÷ inlet coefficient about 0.4 ¢/°F = 75 °F (42 °C) · expansion of the lower reflector, then the core support, inserts the negative reactivity · the drawing unfolds this arithmetic in time (time scale arbitrary)

RVACS - the last heat removal, with no pumps

- RVACS is sized to **decay heat after a day (0.5-0.7 %)**; before that the 333 m³ pool soaks it up - **big inventory + 330 K margin + RVACS**, lose one and it fails
- RVACS capacity scales with vessel area → **it caps module power** - the same structure as core diameter capping the HTGR



RVACS concept and capacity - pool inertia buys the first hours, RVACS takes over after · textbook Fig. 10.3

DESIGN	RATED	RVACS
GE-Hitachi PRISM	840 MWt	0.7 % ≈ 5.9 MW
Rockwell SAFR	900 MWt	0.6 % ≈ 5.4 MW
TerraPower Sodium	840 MWt	external air, undisclosed

RVACS Reactor Vessel Auxiliary Cooling System - air rising by itself in the gap between reactor and guard vessel · **PRISM** GE-Hitachi's small SFR · **SAFR** Rockwell's SFR design

Source This textbook §10.4, Table 10.2, Fig. 10.3 · decay heat falls to 0.5-0.7 % of rated about a day after shutdown · Sodium uses external air cooling, capacity undisclosed

DRACS, IRACS and the 4S decay-heat removal system

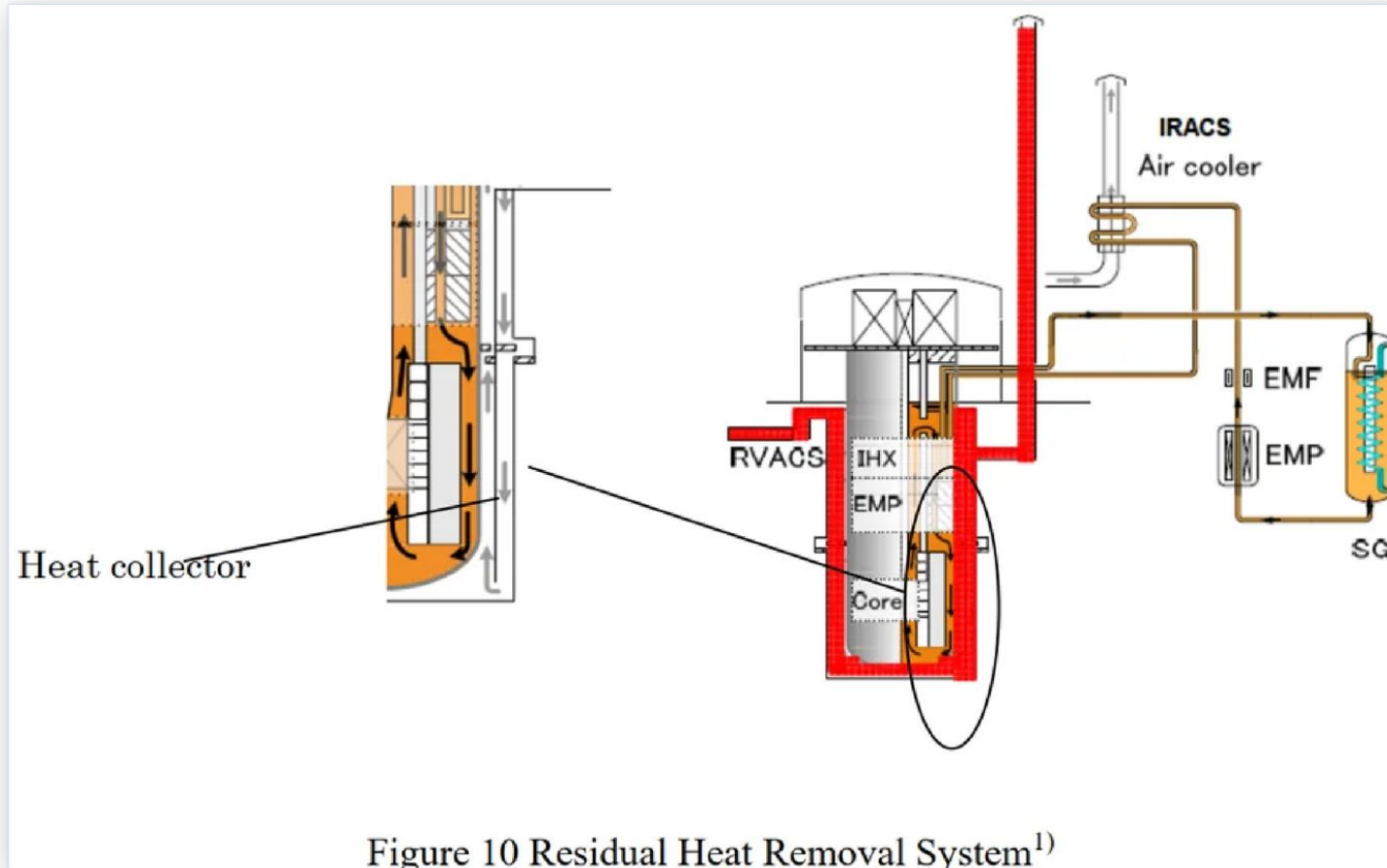


Figure 10 Residual Heat Removal System¹⁾

The 4S decay-heat removal system - an IRACS air cooler on the intermediate loop plus RVACS outside the vessel · textbook Fig. 10.4

RVACS

radiation and convection across the vessel wall. Unpowered, unlimited - but tied to vessel area

DRACS

an exchanger in the pool, with a Na or NaK natural-circulation loop to an air cooler. Not limited by vessel area

IRACS

an air cooler on the intermediate loop. Usually a non-safety first line

DRACS Direct Reactor Auxiliary Cooling - an exchanger dipped in the pool · **IRACS** Intermediate RACS - usually a non-safety first line · **NaK** sodium-potassium alloy, liquid at room temperature

Source This textbook §10.4, Fig. 10.4 · 4S combines IRACS and RVACS

Sodium fires - the actual numbers

- At 11.3 MJ/kg sodium releases less heat than wood (18.6) or petrol (47.8) - it is **a chemistry and aerosol problem**. NaOH fumes corrode, and **200 mg/m³ blocks vision completely**
- Design is driven by **spray fires** - droplets ignite on contact and burn far faster than a pool. Why sodium pipes are double-walled or guarded

QUANTITY	VALUE
Ignition	no fixed value - pool 200-400 °C · intact oxide 275-425 °C · disturbed 200 °C · finely dispersed, spontaneous at room temperature
Pool burn rate · heat flux	2~45 g/(s·m²) · 2.3×10 ⁴ ~ 5.1×10 ⁵ W/m²
Heat of combustion	11.3 MJ/kg (wood 18.6 · petrol 47.8)
Vision blocked	NaOH 200 mg/m³
Response	inert cells (Ar, N₂) · catch pans and honeycomb trays (≤ 30 % open) · limited area · nitrogen injection. No water

spray fire leaked sodium droplets burning in the air · **aerosol** fine particles suspended in air · **honeycomb tray** a catch tray that starves fallen sodium of oxygen

Source This textbook §10.5 · never water · keep cooling for 24 h or more after extinguishing

SFR in defence in depth - where the weight moves, and what remains

- Sodium fires and sodium-water reactions fit no level cleanly - they behave like industrial fire and chemistry. **Why NRC treats fire protection as a separate axis**
- **No commercial reactor has ever validated an accident where large-core positive void worth actually acts (local boiling after a blockage)** - the biggest hurdle in SFR licensing

Level 1 · normal operation

heavier

sodium purity (O under 3 ppm), anti-freeze heating, cover gas and leak monitoring every day

Levels 2-3 · anomalies, accidents

lighter

many pressure scenarios vanish; reactivity events are caught by structural feedback

A new axis · fire and chemistry

appears

sodium fires and sodium-water reactions - industrial fire and chemistry, not reactor accidents

defence in depth independent layered protection · flow blockage an assembly channel becoming blocked · SAS4A/SASSYS-1 ANL's SFR safety codes

Source This textbook §10.6 · the five-level frame is from *SMR Design Requirements and ESF* · integral data for accidents where positive void worth comes into play date from the 1980s only

11 Running a system code on sodium

Sodium is a liquid with no phase change and almost no compressibility - the simplest property class of the three. The difficulty comes from elsewhere: correlations, axial conduction, pool stratification, long integration times.

Four difficulties of a sodium system code

- Easier than helium, harder in places - the properties are simplest, but new problems appear in **closures, numerics and spatial dimension**

1 Correlations must change

2.6 × wrong otherwise (§7.3)

2 Axial conduction matters

the term LWR codes drop - alive in natural circulation

3 Pool stratification dominates

a 1-D network cannot solve it

4 Long integration times

hours of accident, ms-s steps - ratio 10^7

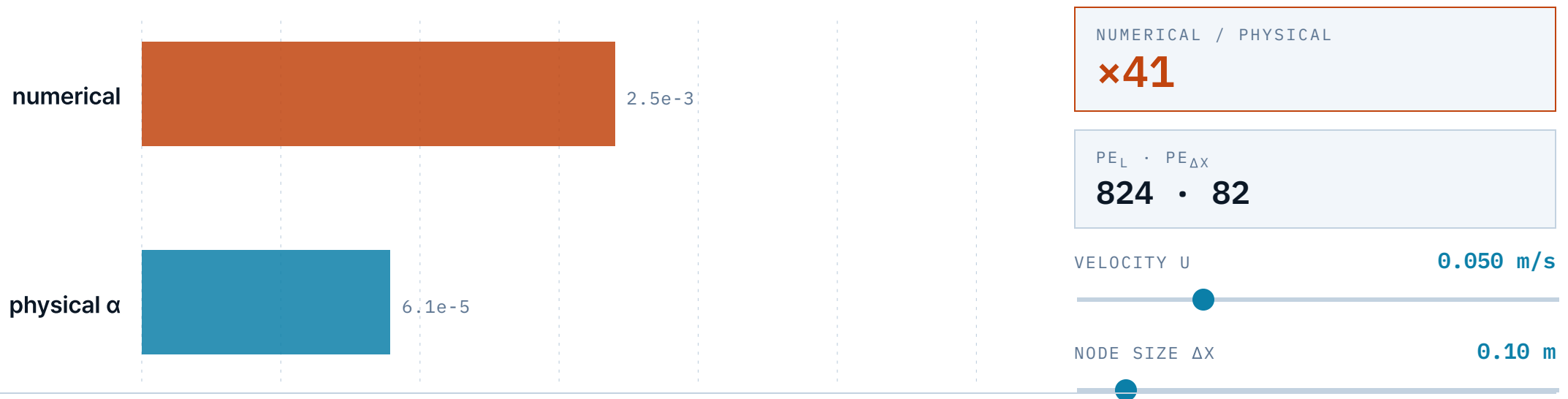
axial conduction conduction along the flow - usually dropped in LWR codes · **thermal stratification** hot fluid layering over cold · **integration time** the total length of the event simulated

Source This textbook §11.1 · no steam tables, flow-regime maps or critical-flow models needed

Axial conduction and numerical diffusion

- When natural circulation drops the velocity to 0.05 m/s, $Pe_L \approx 800$ and $Pe_{\Delta x} \approx 80$ at $\Delta x = 0.1$ m - and donor-cell numerical diffusion sits on top at **about 40 times physical conduction**
- If a thermal front looks smeared, check **numerical diffusion before physics** - finer nodes (costly) or second-order upwinding (a code change). Volume III is first-order donor cell

$$\text{convection/conduction} \sim \rho u c_p L / k = Pe_L \quad D_{\text{num}} \approx (u \Delta x / 2) (1 - Co)$$



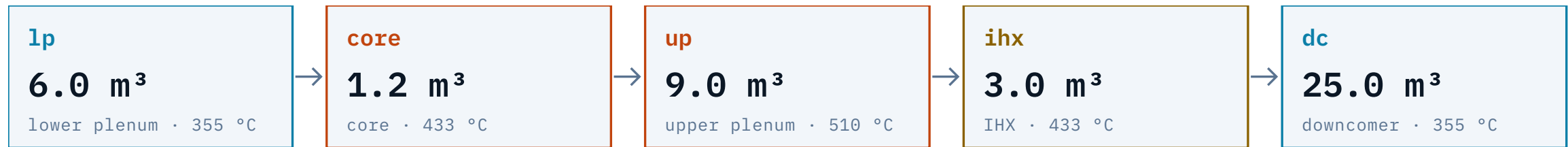
donor cell first-order upwind differencing · numerical diffusion discretisation error acting like diffusion and smearing fronts · Co Courant number $u\Delta t/\Delta x$ · thermal front a sharp temperature change

diffusivity m^2/s (log)

Calculation This textbook §11.2 · sodium at 500 °C ρ 835, c_p 1264, k 64.2 $\rightarrow \alpha = 6.1 \times 10^{-5} m^2/s$ · $Pe_L = \rho u c_p L / k$ ($L = 1$ m) · numerical diffusion $\approx u\Delta x/2 \cdot (1 - Co)$, shown at its $Co \rightarrow 0$ maximum · an LWR at 5 m/s has $Pe_L \approx 3 \times 10^7$

A five-volume model of 4S, and the heat capacity of the pool

- The core is 1.2 m³, the pool over 40 m³ - with no heat removal at all it warms **about 25 K an hour**. That is the arithmetic behind "hours of grace"
- Pressure waves want 0.4 ms, the event lasts hours - a ratio of **10⁷**. ① semi-implicit ($\Delta t \leq 167$ ms at 6 m/s) ② up to 20 s in natural circulation (adaptive) ③ solve the steady state separately



pool temperature rise in an hour with no heat removal · 1 % decay heat

sodium pool (43.5 MJ/K)

≈ 25 K

44.2 m³ in all, of which the core is 1.2 m³ - the volume ratio is the thermal inertia

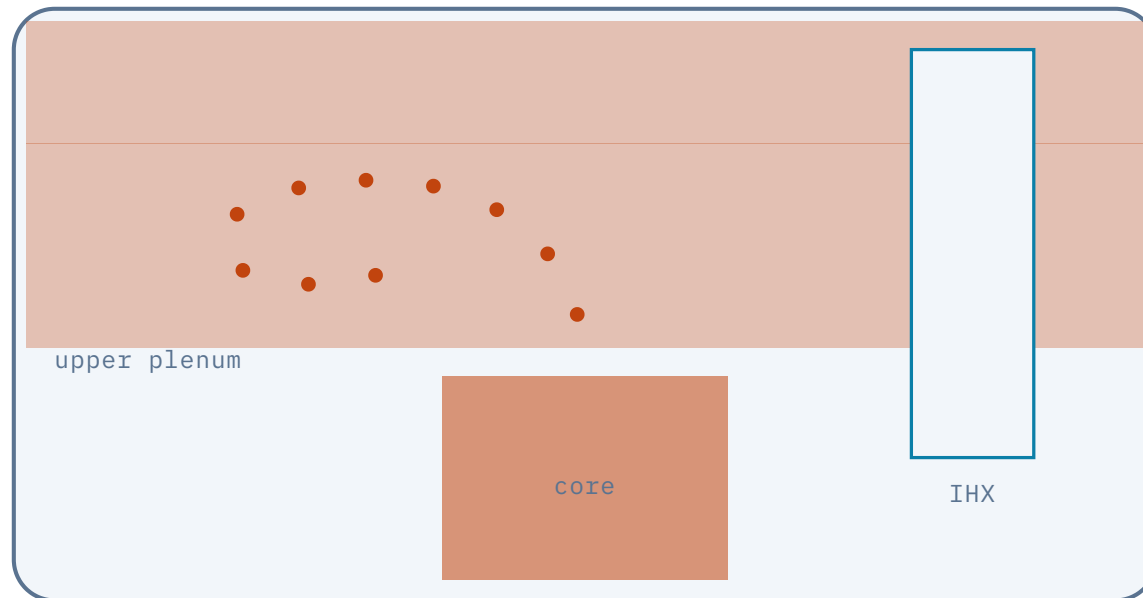
note - T in DenseLiquid(1014, 0.235) is in kelvin

DenseLiquid Volume III's near-incompressible liquid class · **adaptive time step** Δt grown and shrunk automatically with the state · **steady-state solver** finds the steady solution without marching in time

Calculation This textbook §11.3, `sfr_loop.py` · `DenseLiquid(1014, 0.235 - T in kelvin)` · pool about 40 m³ × 856 kg/m³ × 1270 J/(kg·K) = 43.5 MJ/K · decay heat an hour after shutdown about 300 kW (1 % of 30 MWt) → 0.007 K/s, about 25 K/h · pressure-wave $\Delta t \approx 0.4$ ms

What the Volume III code cannot solve - upper-plenum stratification

- When flow falls, hot sodium layers over cold and does not mix → **IHX inlet temperature** is nothing like predicted, **the vessel wall sees thermal stress**, and **the natural-circulation drive** changes
- Finer volumes can mimic it, but the inter-layer mixing coefficient **cannot be set without experiments** - know that this is where our code stops



WHAT A 1-D VOLUME ASSUMES

holds - one temperature

THINK Lab link - RIM-PIV in a transparent fluid at matched Re and Ri gives exactly the data to set this mixing coefficient.

Forced flow - mixed

Loss of flow - stratified

perfect mixing one temperature per control volume · **thermal stress** stress from uneven expansion · **SAM** ANL's MOOSE-based system code

Source This textbook §11.4 · 1-D control volumes assume perfect mixing · ANL's SAM advertises "computationally efficient multi-dimensional flow models for mixing and stratification in large enclosures" · the drawing is a concept sketch

SFR system codes - does it need to handle boiling?

- The choice turns on **whether boiling must be handled** - single-phase system transients can use the RELAP5 extension; **boiling and severe accidents need SAS4A/SASSYS-1**

CODE	WHAT IT CONTAINS	USE
SAS4A/SASSYS-1 (ANL)	multi-slug and bubble sodium boiling · fuel and cladding expansion and deformation · molten fuel · point kinetics with feedback. SAS4A inside the core and damage; SASSYS-1 the plant outside and loss of heat sink	the reference SFR safety code · 1960s lineage
SAM (ANL, MOOSE)	1-D finite-element network + multi-D mixing in large spaces + multi-channel assemblies + point kinetics. Couples to MAMMOTH, Proteus, Nek5000, BISON, TRACE	adopted by NRC for SFRs · also LFR, MSR, gas, heat-pipe
RELAP5 / RELAP5-3D	sodium is not a native fluid. 2026 Pisa liquid-metal equation of state (linearised specific volume, within 1 % for sodium)	single-phase transients only - “not for boiling-dominated transients” (authors’ warning)

Integral data for validating SFR system codes

- The EBR-II SHRT benchmark makes an excellent graduate project - put in the coast-down and a linear structural-feedback term and watch power fall by itself. Seeing ch. 10 come alive in code is this book’s goal
- There are no integral data for accidents where positive void worth acts - choose a topic knowing licensing stalls here and the gap is visible

DATA	WHAT IT VALIDATES
EBR-II SHRT-45R (1986)	loss of flow without scram · structural feedback · transition to natural circulation
EBR-II SHRT-17 and others	natural-circulation flow · upper-plenum temperature
Phénix natural circulation (1990s)	large-pool natural circulation
Phénix end-of-life tests (2009)	core behaviour · plant transients
FFTF transient tests	loop-type behaviour

Six numbers to keep from Part 2

- Sodium carries heat without moderating, metal fuel makes the core one thermometer, and the pool buys time - **these six numbers are the evidence**

Pr 0.005

a water correlation is 2.6 × off

collapse onto Pe - Mikityuk

650 °C

what sets the design temperature

the FCCI eutectic, not 883 °C boiling

124 K

metal fuel stays tied to the coolant

oxide 796 K (25 kW/m)

75 %

smear density

bubbles link at 30 % swelling

75 °F

where the 1986 LOHSWS stopped

30 ¢ ÷ 0.4 ¢/°F

0.5~0.7 %

RVACS = decay heat after a day

before that, the pool absorbs it

Abbreviations (1 of 2) - reactors, fuel, core

Abbreviation	Expansion	What it means
SFR · LFR	Sodium- / Lead-cooled Fast Reactor	sodium- and lead-cooled fast reactors
4S	Super Safe, Small and Simple	Toshiba's 30 MWt pool SFR, 30 years unrefuelled
EBR-II	Experimental Breeder Reactor II	US 62.5 MWt pool - the 1986 tests
FFTF	Fast Flux Test Facility	US loop-type fast test reactor
PRISM	Power Reactor Innovative Small Module	GE-Hitachi's small SFR
PGSFR	Prototype Gen-IV SFR	Korea's 150 MWe prototype - R&D ended 2020
MOX	Mixed Oxide fuel	mixed U-Pu oxide fuel
U-Pu-Zr	—	metal fuel alloy - solidus 1077 °C
HT-9 · ODS	—	ferritic-martensitic and ODS cladding
FCCI	Fuel-Cladding Chemical Interaction	eutectic at 650-725 °C
CFV	Cœur à Faible effet de Vide	low-void core (ASTRID)
MA	Minor Actinides	Np, Am, Cm
β_{eff}	Effective delayed-neutron fraction	the reference for the dollar
TOP	Transient Overpower	metal fuel has 4-4.5 × margin
at. %	Atom percent burnup	burnup in atom percent

Abbreviations (2 of 2) - systems, safety, codes

Abbreviation	Expansion	What it means
IHX	Intermediate Heat Exchanger	between primary and intermediate sodium
EMP	Electromagnetic Pump	$F = J \times B$, no inertia
RVACS	Reactor Vessel Auxiliary Cooling System	air outside the vessel - unpowered, unlimited
DRACS	Direct Reactor Auxiliary Cooling System	an exchanger in the pool + natural-circulation loop
IRACS	Intermediate RACS	air cooler on the intermediate loop - usually non-safety
LOFWS · ULOF	Loss of Flow Without Scram	the first 1986 test
LOHSWS	Loss of Heat Sink Without Scram	the second 1986 test
SBO	Station Blackout	loss of all AC power
SHRT	Shutdown Heat Removal Tests	EBR-II tests - the validation benchmark
SAS4A/SASSYS-1	—	ANL's reference SFR safety code
SAM	System Analysis Module	ANL's MOOSE-based system code
Na-24	—	activated sodium - 15 h, strong gamma
CNSC · VDR	Canadian Nuclear Safety Commission	Canada's regulator and vendor design review
RIM-PIV	Refractive-Index-Matched PIV	THINK Lab's index-matched PIV
Pe · Ri	Péclet · Richardson number	convection/conduction, buoyancy/inertia

3 April 1986 - a reactor that stopped **without rods**

Every small SFR's "passive safety" is a claim to reproduce what was measured that day.
And no one has yet demonstrated an accident where large-core positive void worth actually acts.

Next is the **MSR** - where the fuel flows, and one assumption nuclear engineering has made for sixty years falls away.